Chapter 4

FIELD-EFFECT TRANSISTORS

CHAPTER OUTLINE

4-1 Structure of Field-Effect Transistors

4-2 JFET Characteristics

4-3 JFET Biasing

4-4 MOSFET Characteristics

4-5 MOSFET Biasing

4-6 FET Linear Amplifiers

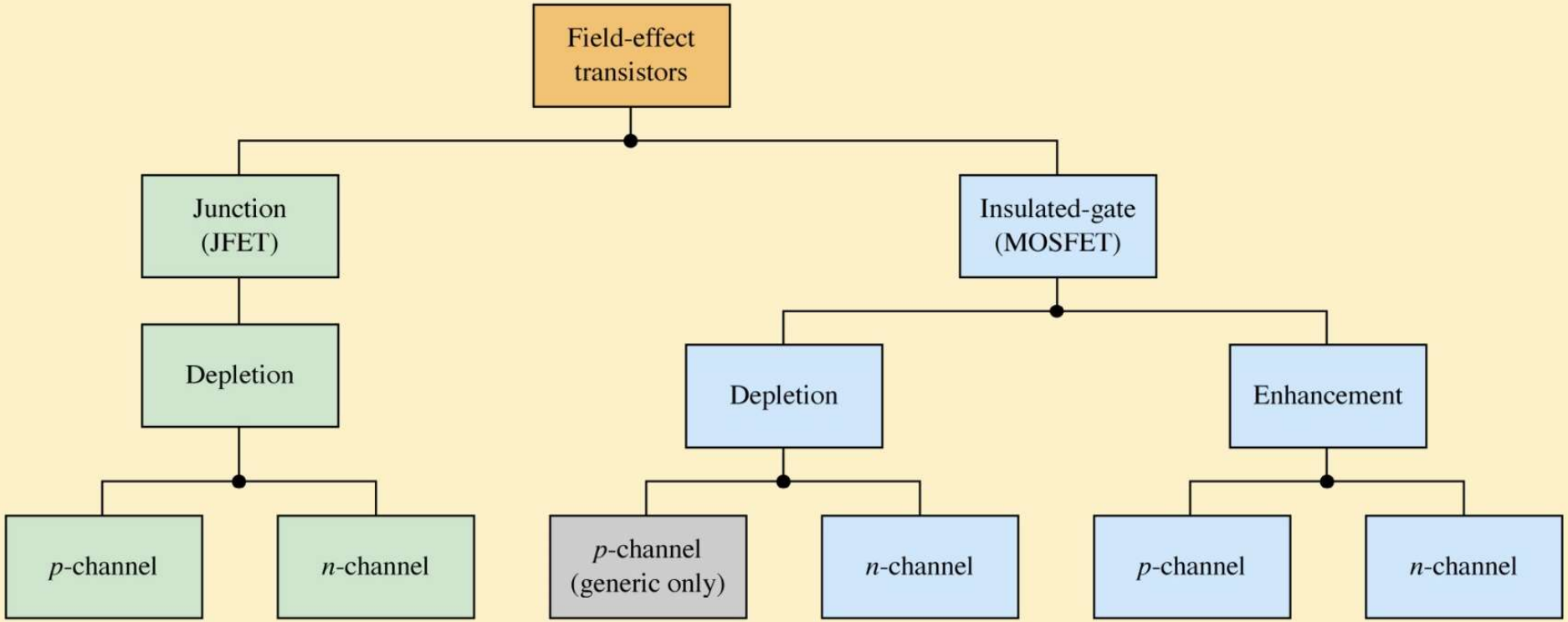
4-7 FET Switching Circuits

4-1 Structure of Field-Effect Transistors

- **Field-Effect Transistor (FET)**
  - In a FET, a narrow conducting channel is connected to two leads called the source and the drain.
  - Conduction in the channel is controlled by an electric field, established by a voltage applied to a third lead called the gate.
- **The FET Family**
  - Junction Field-Effect Transistor (JFET)
  - Metal Oxide Semiconductor Field-Effect Transistor (MOSFET)
- **Characteristics of FETs**
  - Very high input resistance and low electrical noise
  - Lower gain and less linear than BJTs

4-1 Structure of Field-Effect Transistors

- **Classification of FETs**
  - A characteristic that differs between various types of FETs is their dc behavior



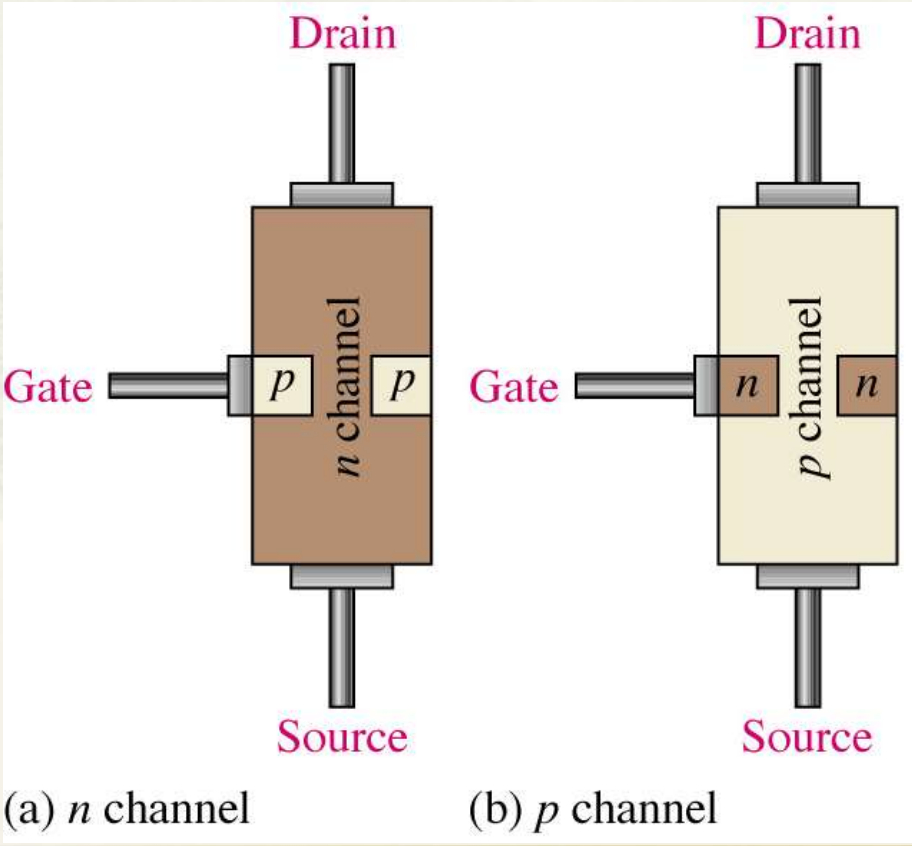
4-2 JFET Characteristics

- JFET Operation
- JFET Symbols
- Drain Characteristic Curve
- Pinch-Off Voltage
- V_{GS}Controls I_D
- Cutoff Voltage
- Comparison of Pinch-Off and Cutoff
- JFET Transconductance Curve
- JFET Input Resistance

JFET Operation

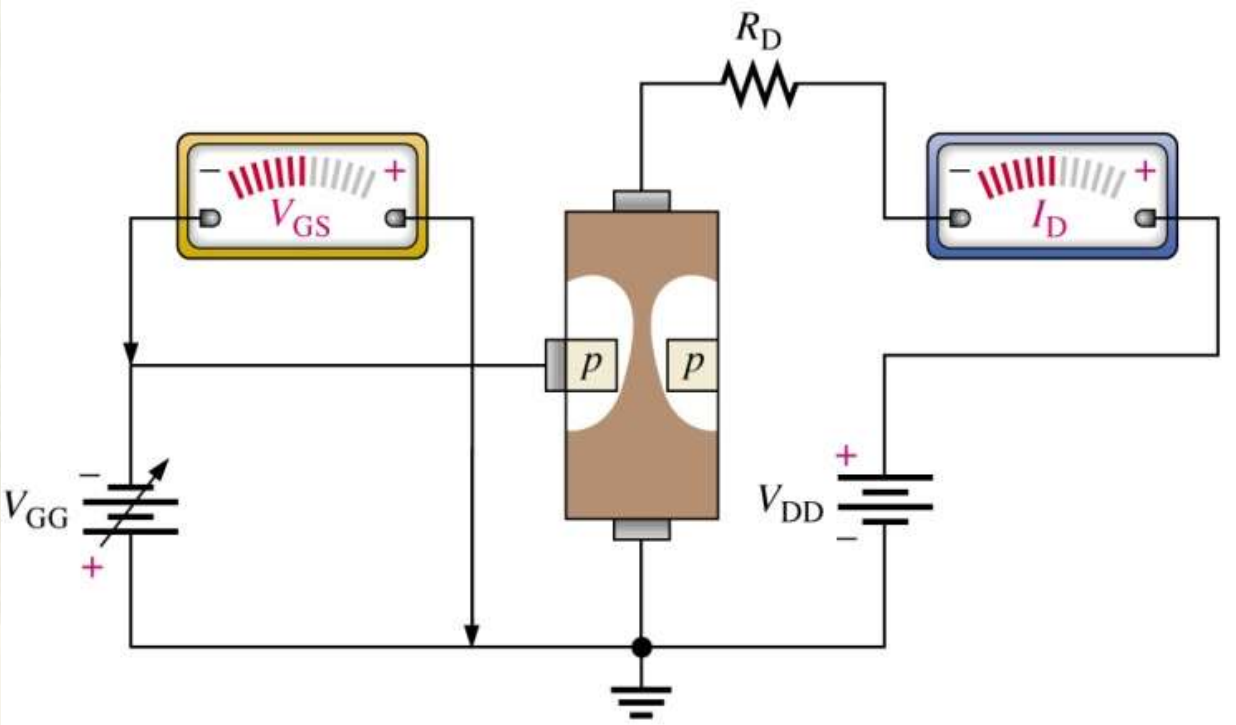
- **Junction Field-Effect Transistor (JFET)**
  - Three leads: source, drain, gate
  - Types:
    - ✓ N-channel JFET: electrons as carriers
    - ✓ P-channel JFET: holes as carriers

With no external voltages, the channel can conduct current in either direction.



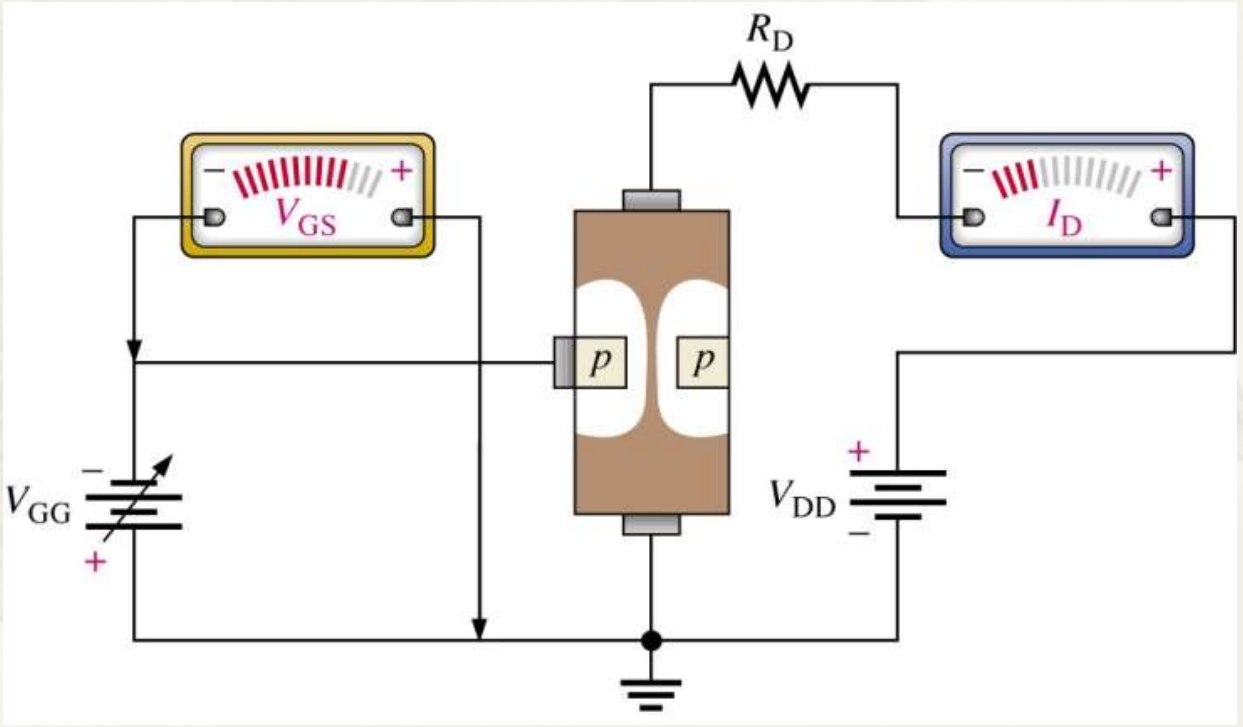
JFET Operation

- **Effects of V_{GG} on channel width, resistance, and drain current**
  - Reverse-biasing of the gate-source junction
  - JFET biased for conduction



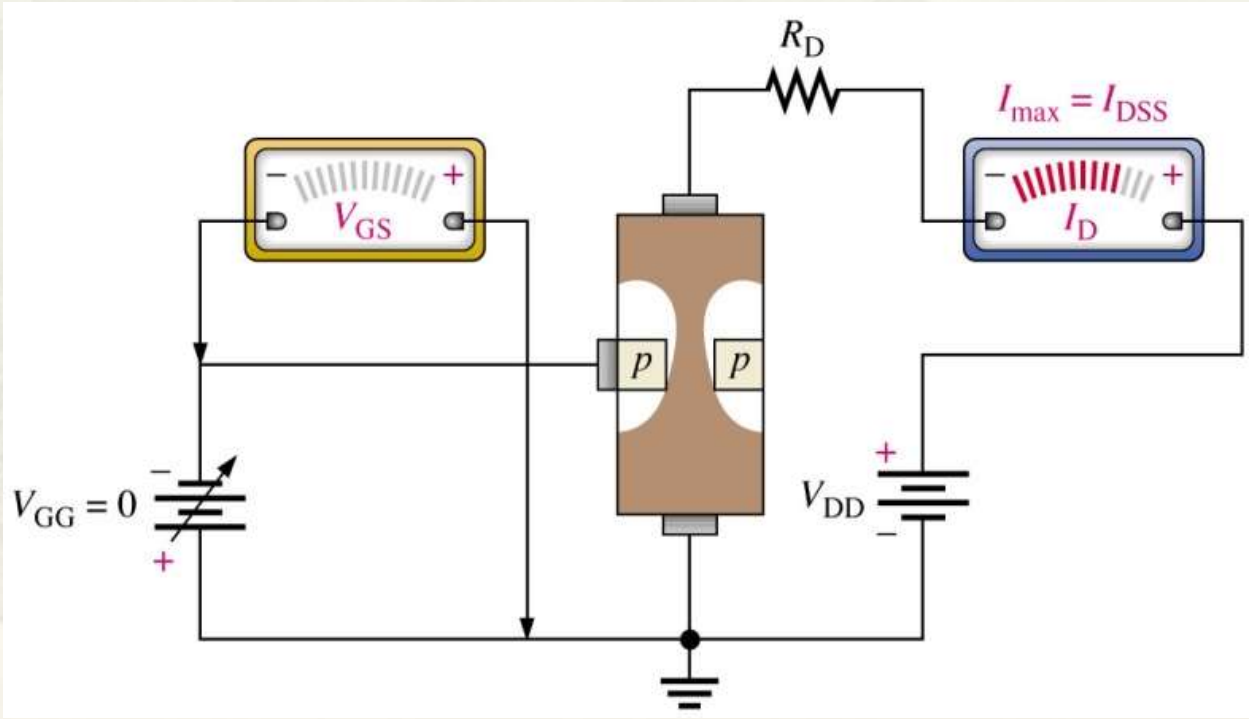
JFET Operation

- **Effects of V_{GG} on channel width, resistance, and drain current**
  - Reverse-biasing of the gate-source junction
  - JFET biased for conduction
  - Greater V_{GG} narrows the channel



JFET Operation

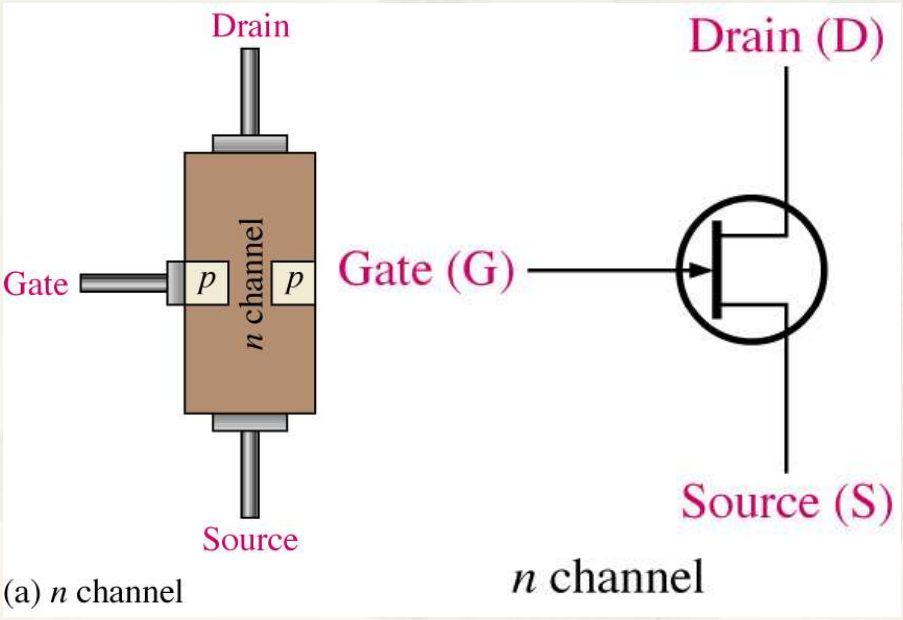
- **Effects of V_{GG} on channel width, resistance, and drain current**
  - Reverse-biasing of the gate-source junction
  - JFET biased for conduction
  - Greater V_{GG} narrows the channel
  - Less V_{GG} widens the channel



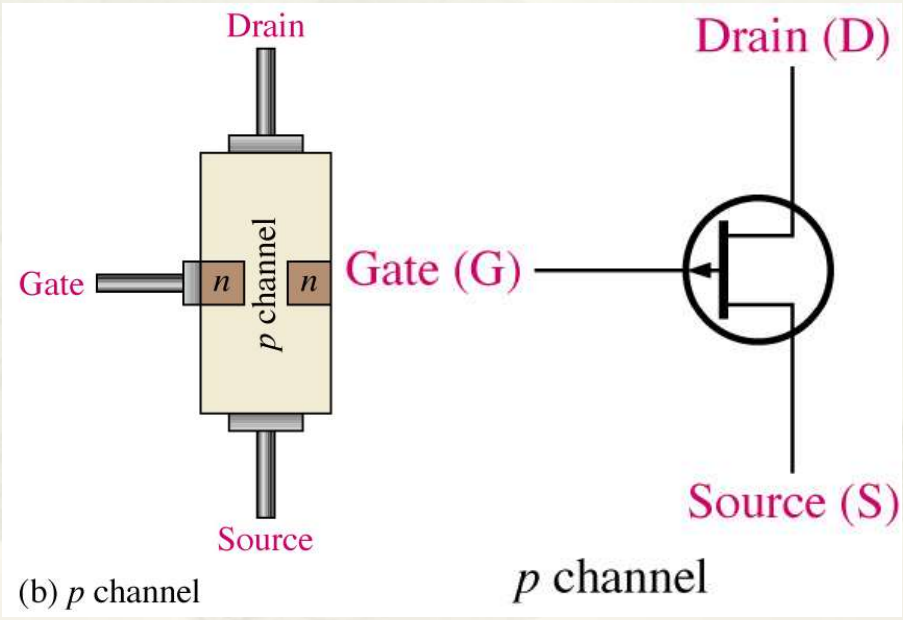
The key idea is that the channel width is controlled by the gate voltage.

JFET Symbols

N-channel FET

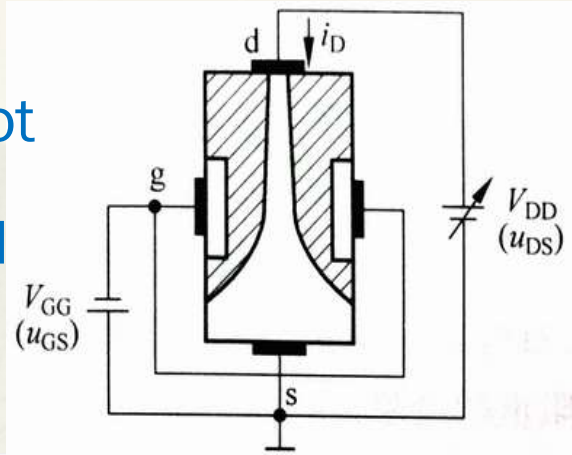


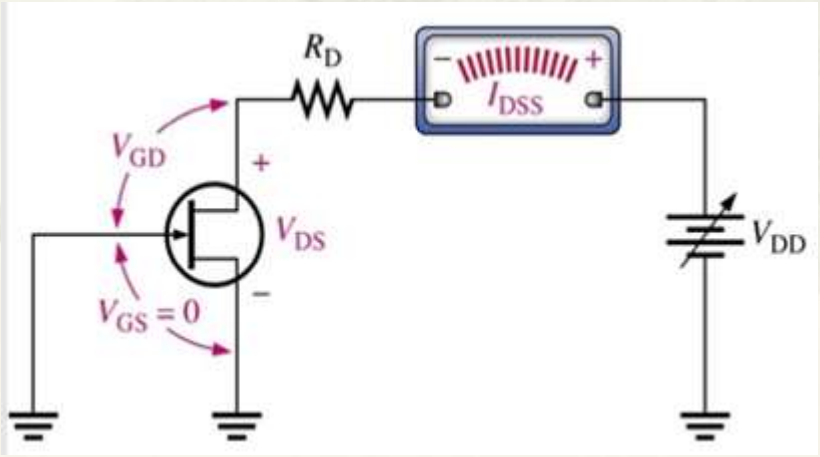
P-channel FET

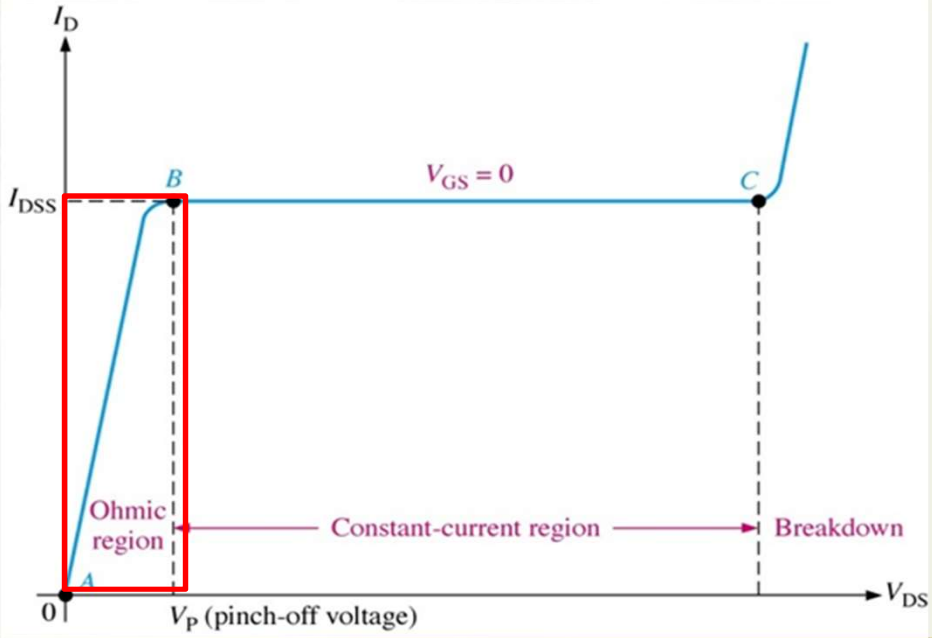


Drain Characteristic Curve

- **Ohmic region**
  - The channel resistance is essentially constant because the depletion region is not large enough to have a significant effect.
  - The value of the resistance can be changed by the gate voltage.
  - Possible to be used as a voltage-controlled resistor.

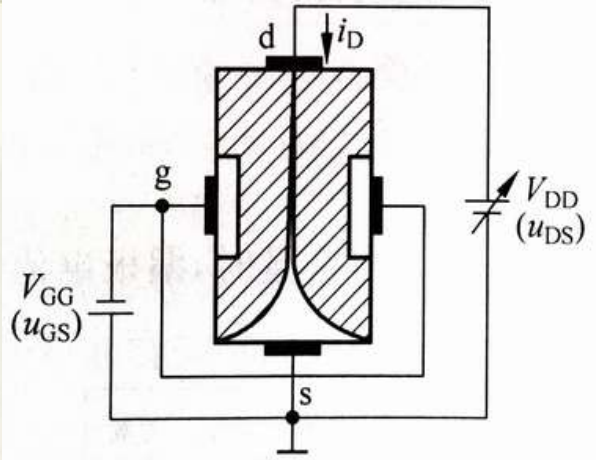


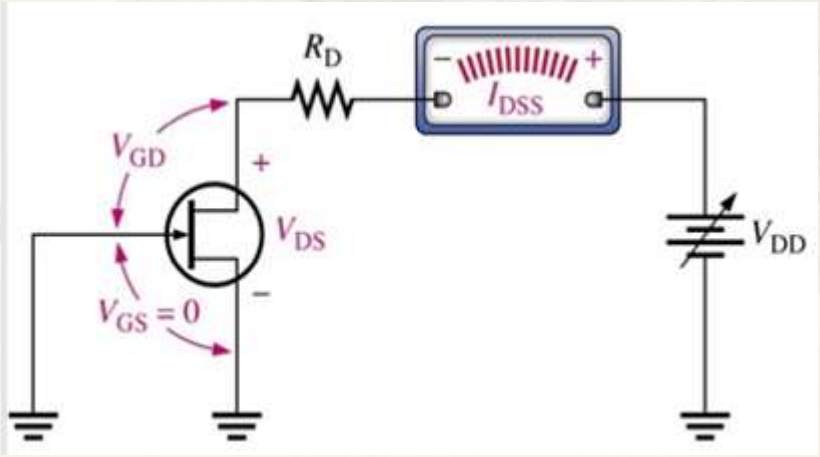


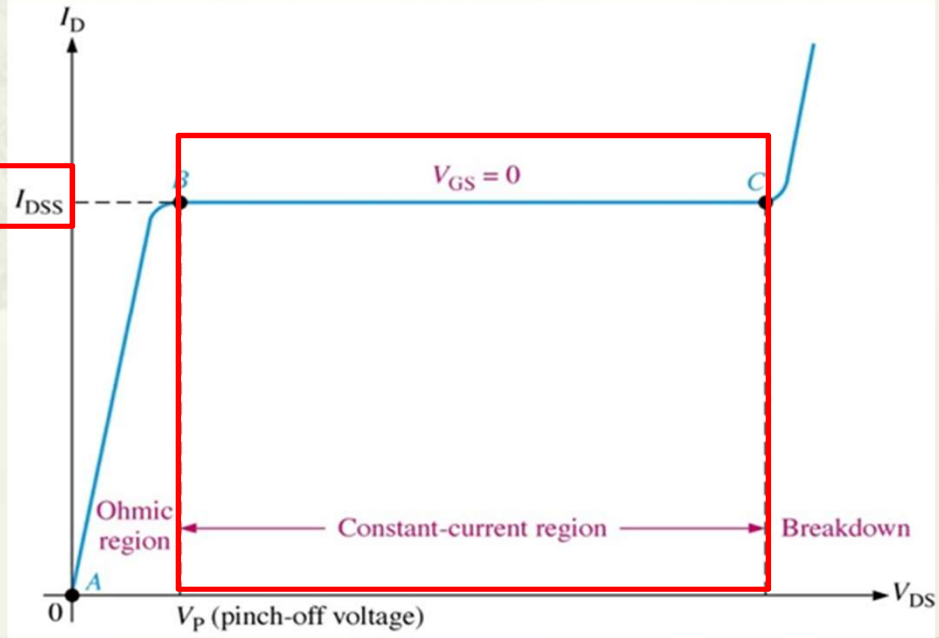


Drain Characteristic Curve

- **Constant-current region**
  - The reverse-bias voltage from gate to drain (V_{GD}) produces a depletion region large enough to offset the increase in V_{DS}, thus keeping I_D relatively constant.
  - I_{DSS}: Drain to source current with gate shorted

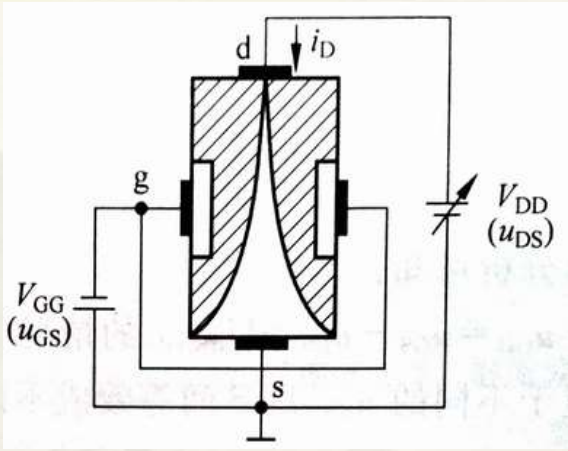


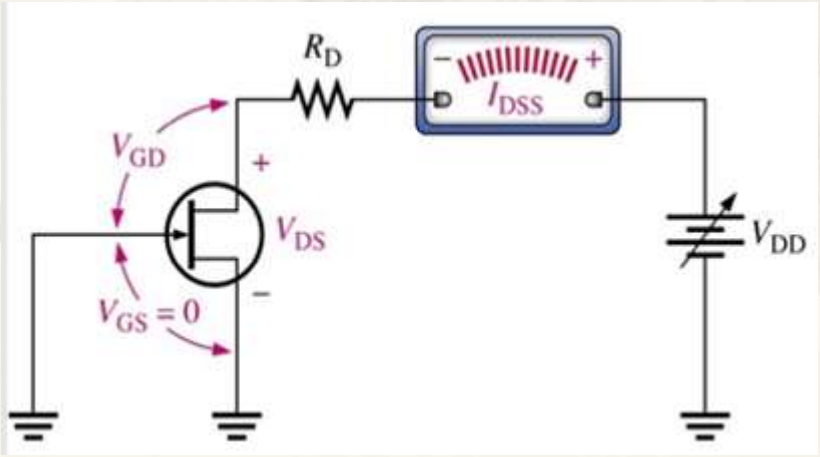


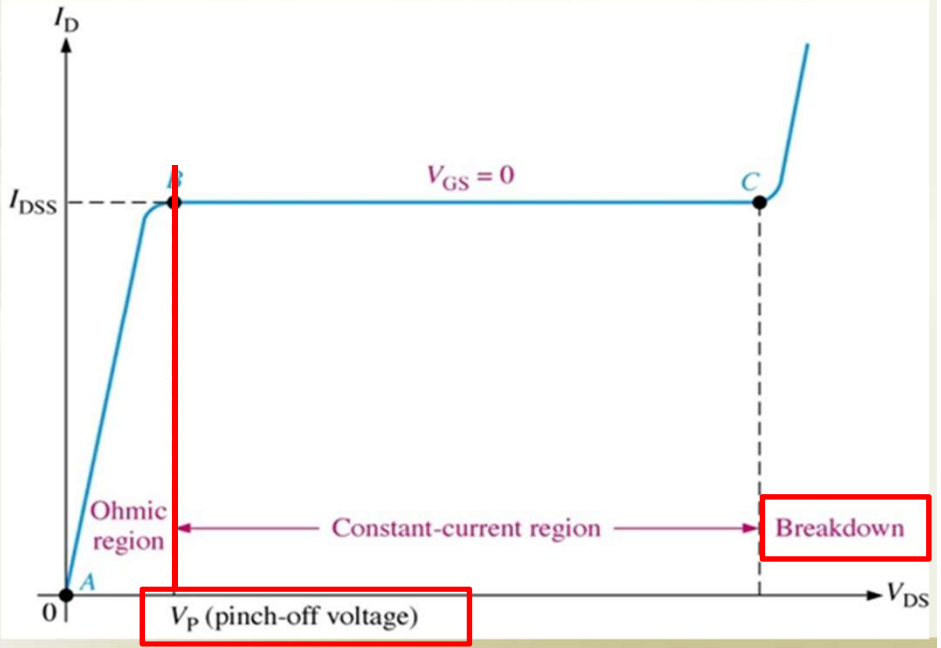


Pinch-Off Voltage

- **Pinch-off voltage**
  - The value of V_{DS} at which I_D becomes essentially constant is the pinch-off voltage, V_P.
  - For a given JFET, V_P has a fixed value.

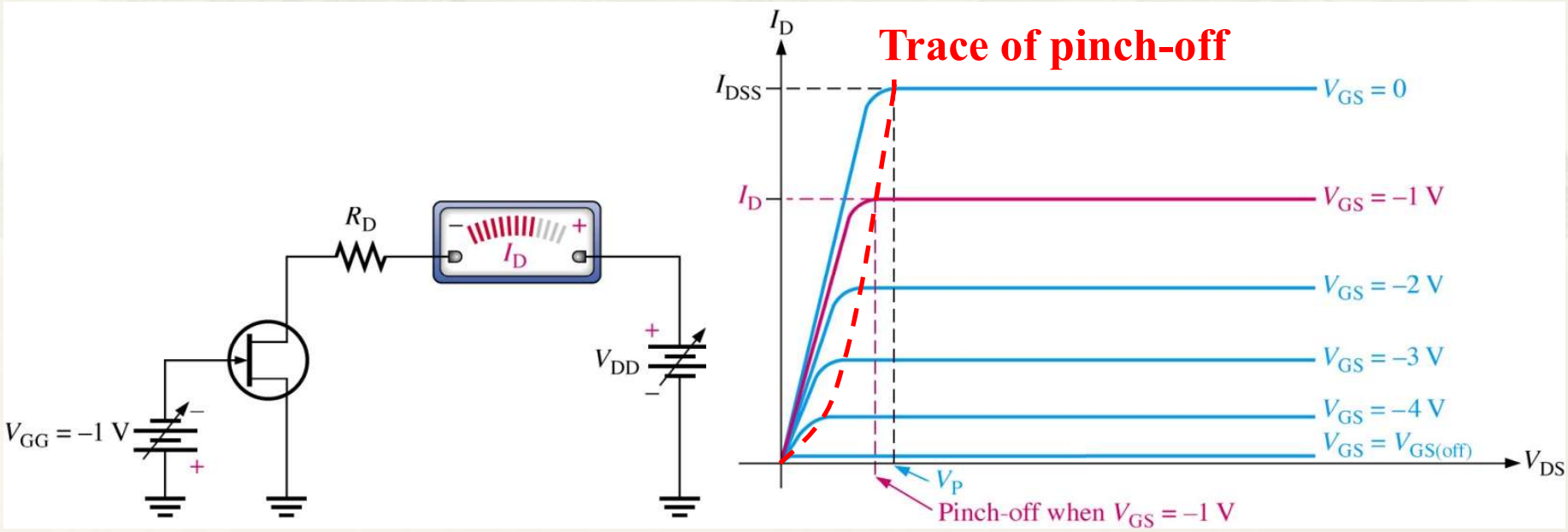






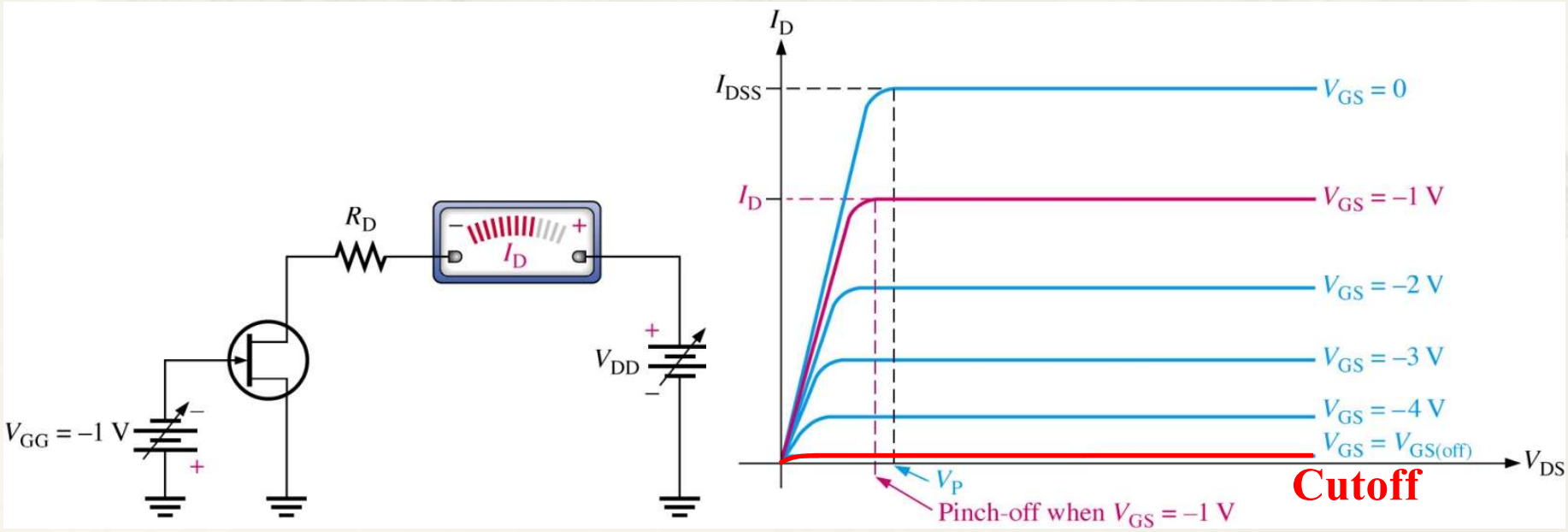
V_{GS} Controls I_D

- I_D decreases as the magnitude of V_{GS} is increased to larger negative values because of the narrowing of the channel.
- The amount of drain current is controlled by V_{GS}



Cutoff Voltage

- **Cutoff voltage**
  - The value of V_{GS} that makes I_D approximately zero is the cutoff voltage, $V_{GS(off)}$.
  - The cutoff effect is caused by the widening of the depletion region to a point where it completely closes the channel.



Comparison of Pinch-Off and Cutoff

Pinch-off Voltage – V_P

- Represents the drain-to-source voltage at which the drain current becomes constant when $V_{GS} = 0\text{V}$
- Voltage from drain to source
- Positive for an n-channel JFET

Cutoff Voltage – $V_{GS(off)}$

- Represents the negative gate-to-source voltage that reduces the drain current to zero.
- Voltage from gate to source
- Negative for an n-channel JFET

- Always equal in magnitude but opposite in sign.
- $V_P = -V_{GS(off)}$

Example 4-1

Problem

For the *n*-channel JFET in Figure 4–7, $V_{GS(off)} = -4$ V and $I_{DSS} = 12$ mA. Determine the *minimum* value of V_{DD} required to put the device in the constant-current region of operation.

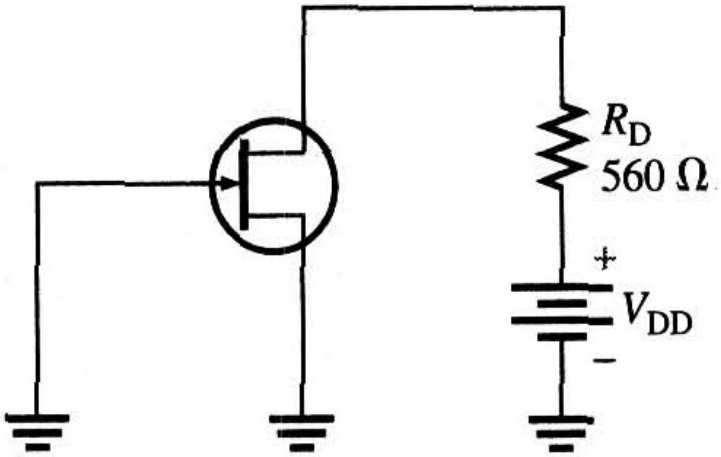


FIGURE 4–7

Solution

Since $V_{GS(off)} = -4$ V, $V_P = 4$ V. The minimum value of V_{DS} for the JFET to be in its constant-current region is

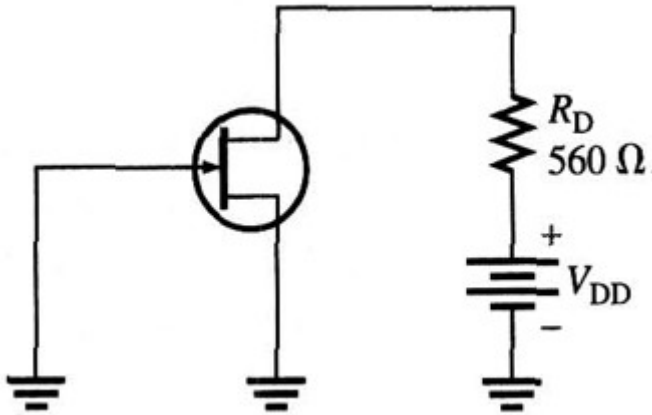
$$V_{DS} = V_P = 4 \text{ V}$$

Example 4-1

Solution

In the constant-current region with $V_{GS} = 0$ V,

$$I_D = I_{DSS} = 12 \text{ mA}$$



The drop across the drain resistor is

$$V_{R_D} = (12 \text{ mA})(560\ \Omega) = 6.7 \text{ V}$$

Applying Kirchhoff's law around the drain circuit gives

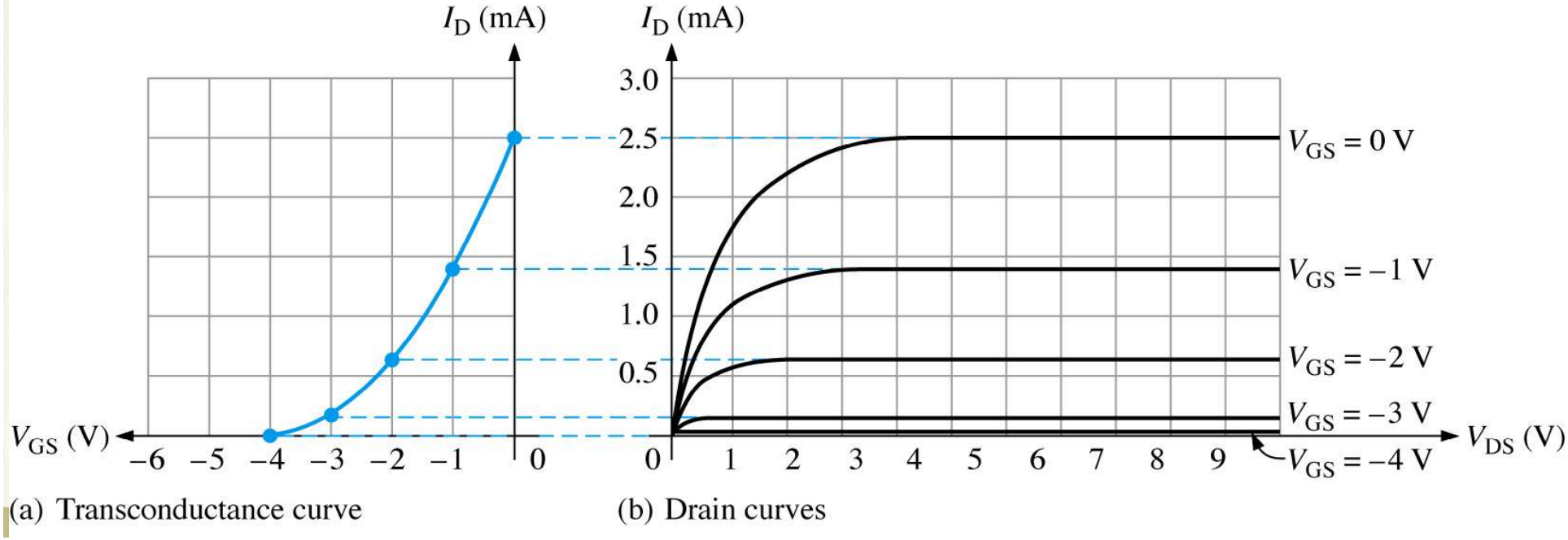
$$V_{DD} = V_{DS} + V_{R_D} = 4 \text{ V} + 6.7 \text{ V} = \mathbf{10.7 \text{ V}}$$

This is the minimum value of V_{DD} to make $V_{DS} = V_P$ and to put the device in the constant-current region.

JFET Transconductance Curve

- **Transfer curve**
  - Show the output for a given input.
- **Transconductance (g_m or y_{fs})**
  - $g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{I_d}{V_{gs}}$

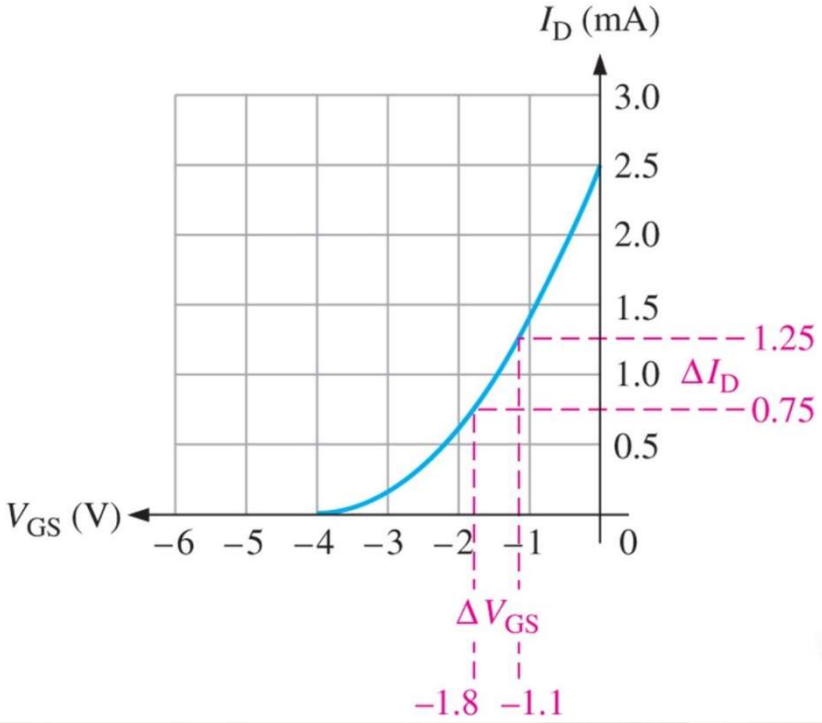
FETs have a nonlinear trans-conductance curve!



Example 4-2

Problem

For the curve in Figure 4–9, determine the transconductance at $I_D = 1.0$ mA.



Solution

Select a small change in I_D and divide it by the corresponding change in V_{GS} at 1.0 mA.

From the graph, the transconductance is

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{1.25\text{ mA} - 0.75\text{ mA}}{-1.1\text{ V} - (-1.8\text{ V})} = \mathbf{0.714\text{ mS}}$$

JFET Input Resistance

- **JFET input resistance**
  - A JFET operates with its gate-source junction reverse-biased; therefore, the input resistance at the gate is very high.
  - JFET data sheets often specify the input resistance by giving a value for the gate reverse current, I_{GSS}
  - $R_{IN} = \left|\frac{V_{GS}}{I_{GSS}}\right|$
  - In a typical application, the total input resistance will include a resistor connected to the gate. The result is a total input resistance in the 1-10MΩ range.

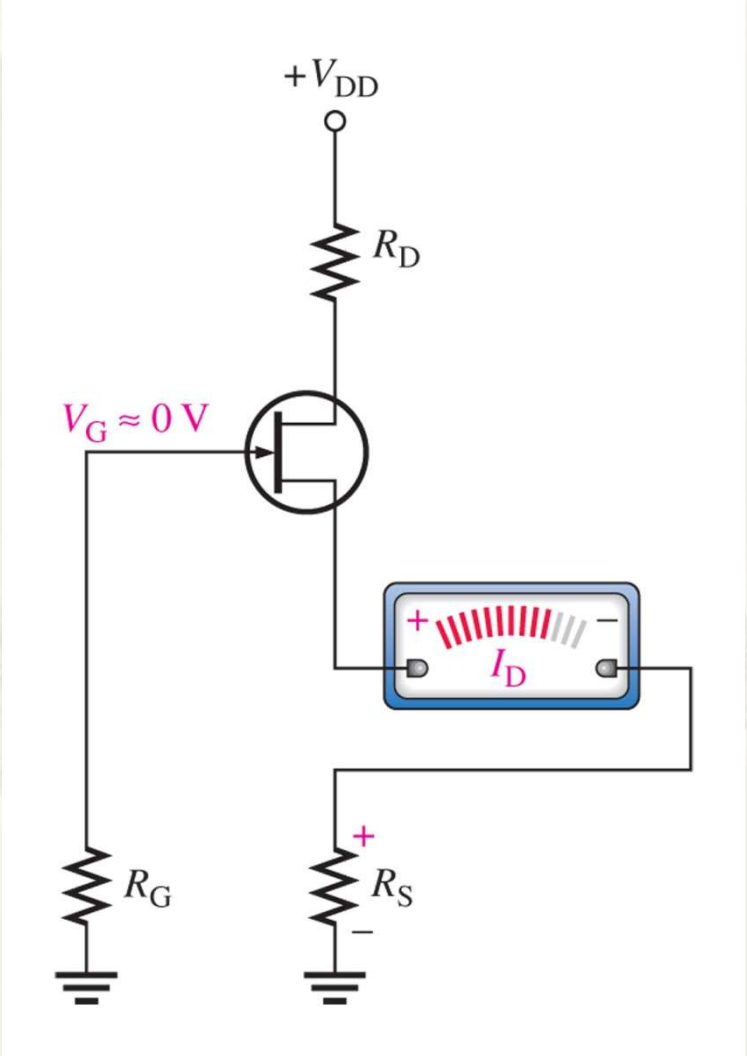
4-3 JFET Biasing

- Self-Biasing a JFET
- Graphical Methods
- Voltage-Divider Bias

The purpose of biasing is to select a proper dc gate-to-source voltage to establish a desired value of drain current.

Self-Biasing a JFET

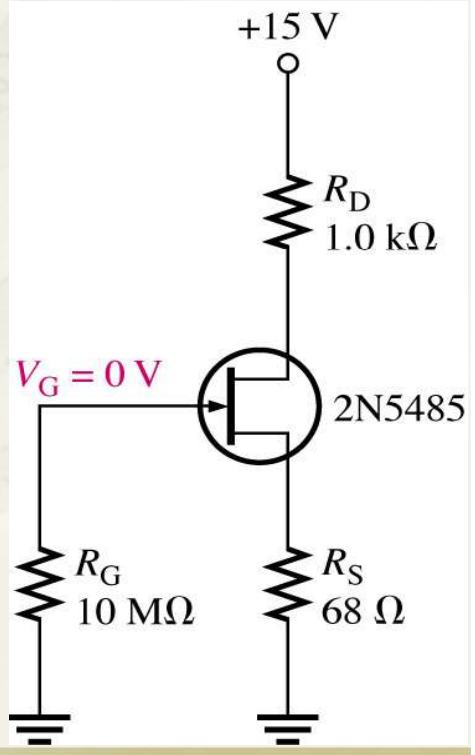
- **Self-bias**
  - Bias the gate at 0V by resistor $R_G$connected to ground, make the source positive with respect to the gate, producing the required reverse bias.
  - $V_{GS} = V_G - V_S = -I_D R_S$
  - $V_D = V_{DD} - I_D R_D$
  - $V_{DS} = V_{DD} - I_D(R_D + R_S)$



Example 4-3

Problem

Find V_{DS} and V_{GS} in Figure 4–11. For the particular JFET in this circuit, assume that the internal parameters are such that a drain current (I_D) of approximately 5.0 mA is produced. Another JFET, even of the same type, may not produce the same results when connected in this circuit due to variations in parameter values.



Example 4-3

Solution

$$V_S = I_D R_S = (5.0\,\text{mA})(68\,\Omega) = 0.34\,\text{V}$$

$$V_D = V_{DD} - I_D R_D = 15\,\text{V} - (5.0\,\text{mA})(1.0\,\text{k}\Omega) = 10.0\,\text{V}$$

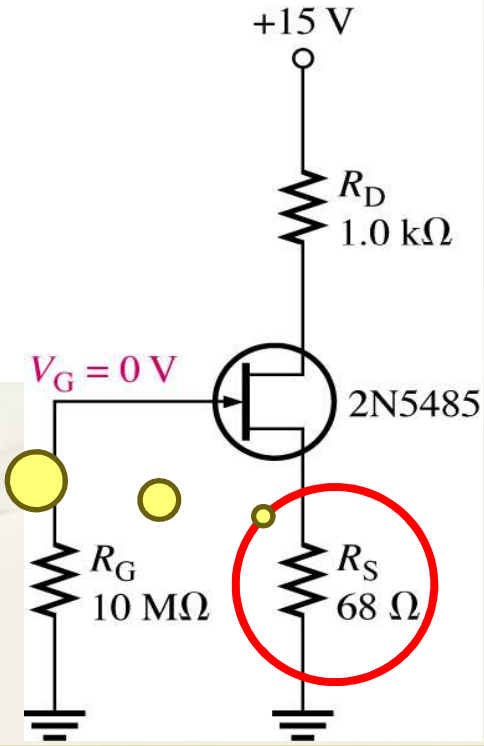
Therefore,

$$V_{DS} = V_D - V_S = 10.0\,\text{V} - 0.34\,\text{V} = \mathbf{9.66\,V}$$

and

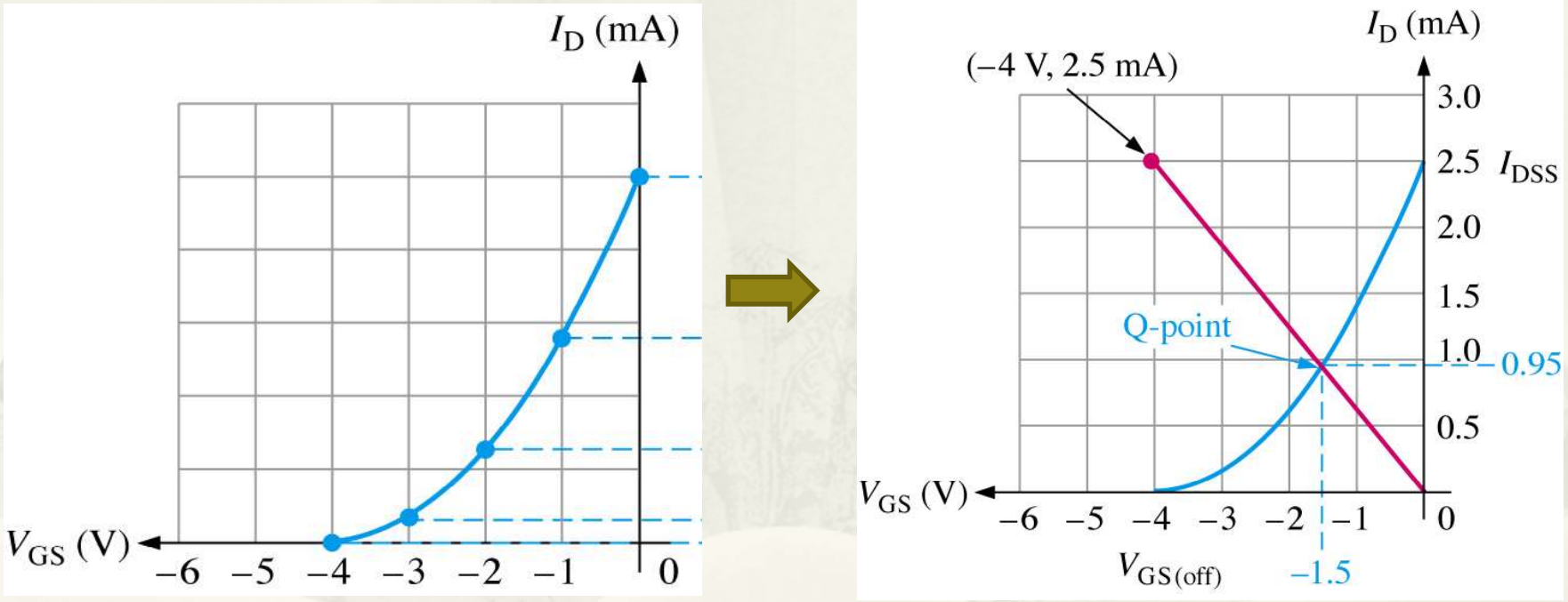
$$V_{GS} = V_G - V_S = 0\,\text{V} - 0.34\,\text{V} = \mathbf{-0.34\,V}$$

How to choose the value?



Graphical Methods

- **A reasonable choice for R_S**

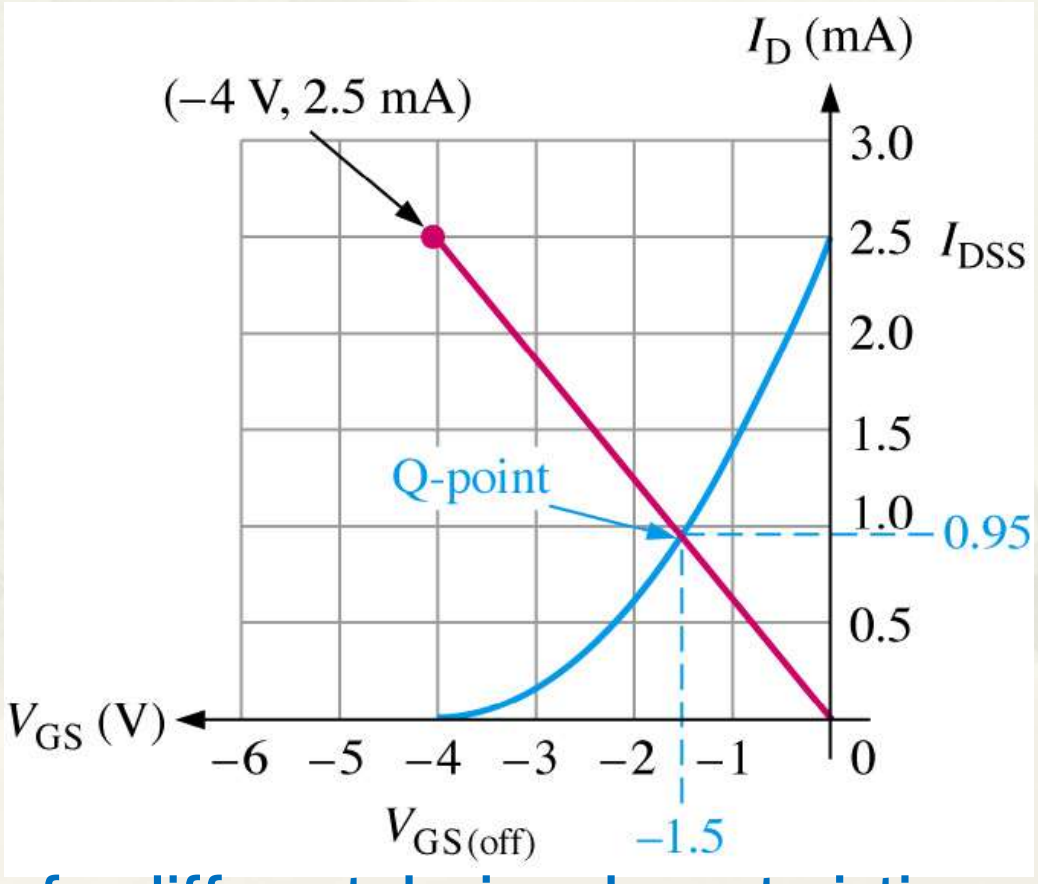


Graphical Methods

- **A reasonable choice for R_S**
  - $R_S = \dfrac{|V_{GS(off)}|}{I_{DSS}}$

$$= \frac{4\ \text{V}}{2.5\ \text{mA}}$$

$$= 1.6\text{k}\Omega$$



- **Q-point**
  - V_{GS}, I_D, V_{DS}

Self-bias helps compensate for different device characteristics between various JFETs.

Example 4-4

Problem

A 2N5457 general-purpose JFET has the following specifications: $I_{DSS(min)} = 1$ mA, $I_{DSS(max)} = 5$ mA, $V_{GS(off)(min)} = -0.5$ V, $V_{GS(off)(max)} = -6$ V. Select a self-bias resistor for this JFET.

Solution

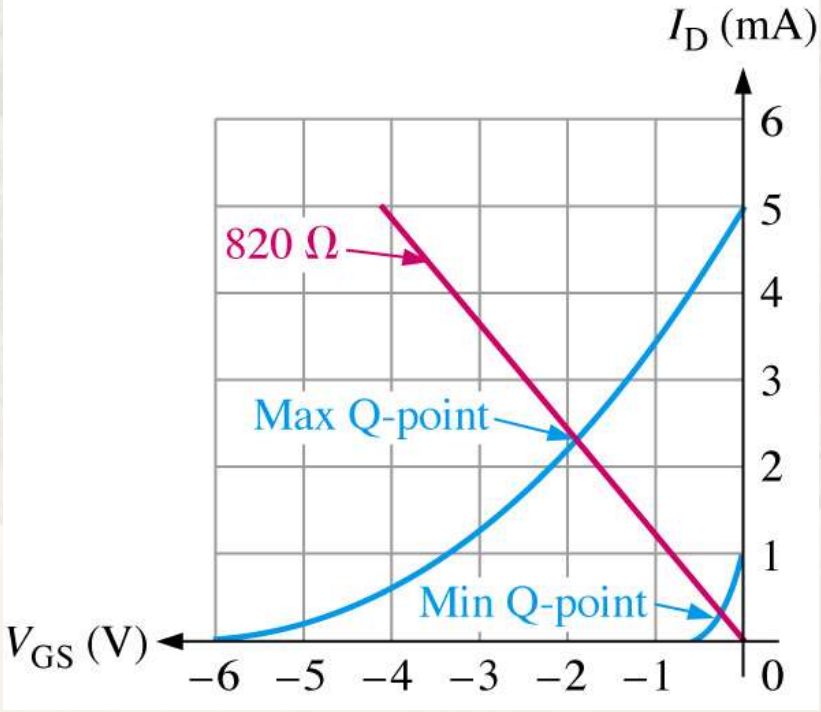
Typical of small-signal JFETs, the range of I_{DSS} and $V_{GS(off)}$ is very large. To select the best resistor, check the extremes of the specified values $V_{GS(off)}$ and I_{DSS}.

$$R_S = \frac{|V_{GS\,(off)(min)}|}{I_{DSS\,(min)}} = \frac{0.5\text{ V}}{1.0\text{ mA}} = 500\ \Omega$$

$$R_S = \frac{|V_{GS\,(off)(max)}|}{I_{DSS\,(max)}} = \frac{6\text{ V}}{5.0\text{ mA}} = 1.2\text{ k}\Omega$$

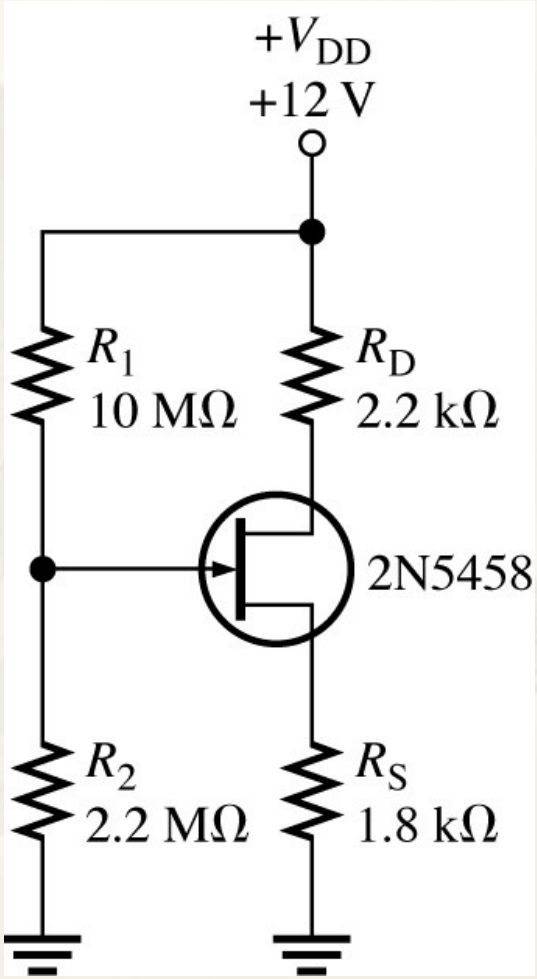
Example 4-4

A good choice is **820 Ω,** a standard value between these extremes. To see what this looks like on the transconductance curve, sketch the curves with this resistor and plot the maximum and minimum Q-points. This is done in Figure 4–13. Despite the extreme variation between the minimum and maximum specification, the 820 Ω resistor represents a good choice for either.



Voltage-Divider Bias

- **Voltage-Divider Bias**
  - More stable with the addition of a voltage divider on the gate circuit
  - Forcing the gate to a positive voltage
  - $V_G = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$



Notice: JFET must still operate with a negative gate-source bias.

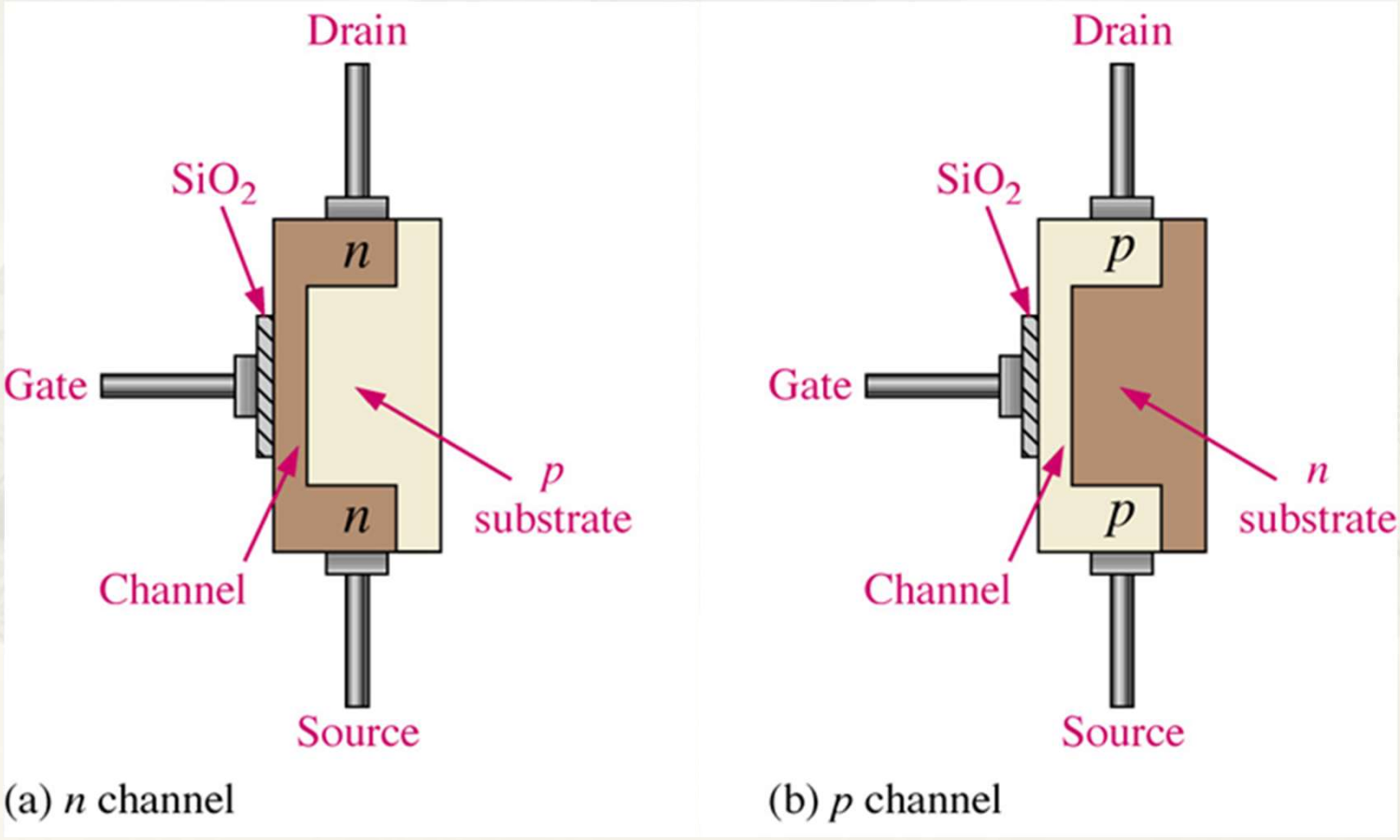
4-4 MOSFET Characteristics

- Depletion MOSFET (D-MOSFET)
- Enhancement MOSFET (E-MOSFET)
- Handling Precautions

The MOSFET differs from the JFET in that it has no *pn* junction structure; instead, the gate of the MOSFET is insulated from the channel by a very thin silicon dioxide (SiO_2) layer.

Depletion MOSFET (D-MOSFET)

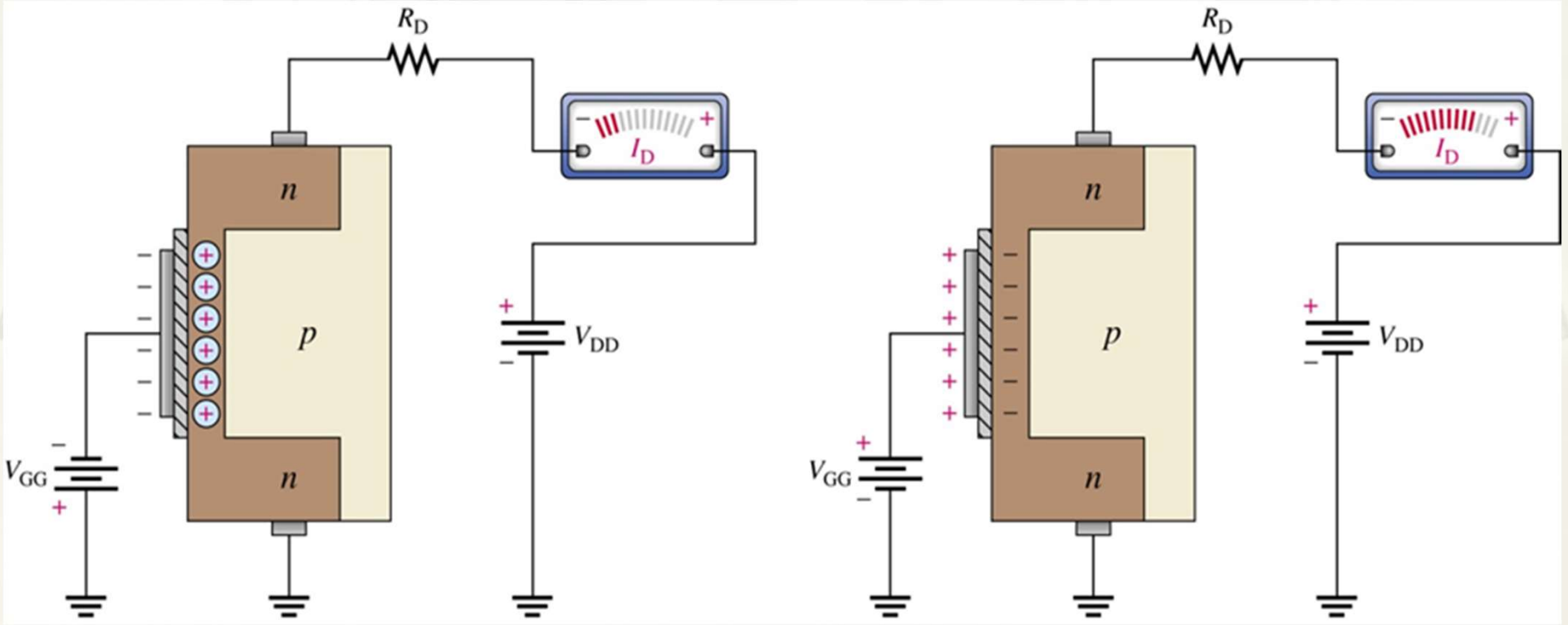
- **Structure of D-MOSFET**



The SiO_2 layer is exaggerated for clarity.

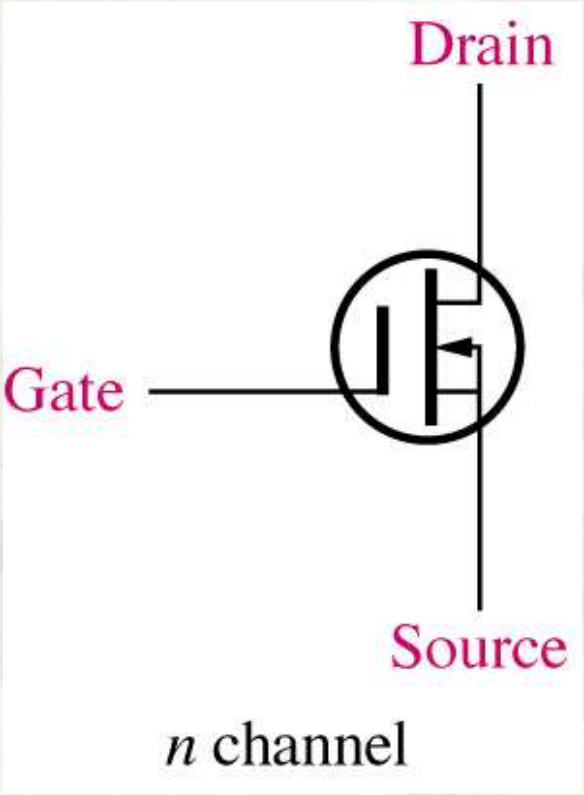
Depletion MOSFET (D-MOSFET)

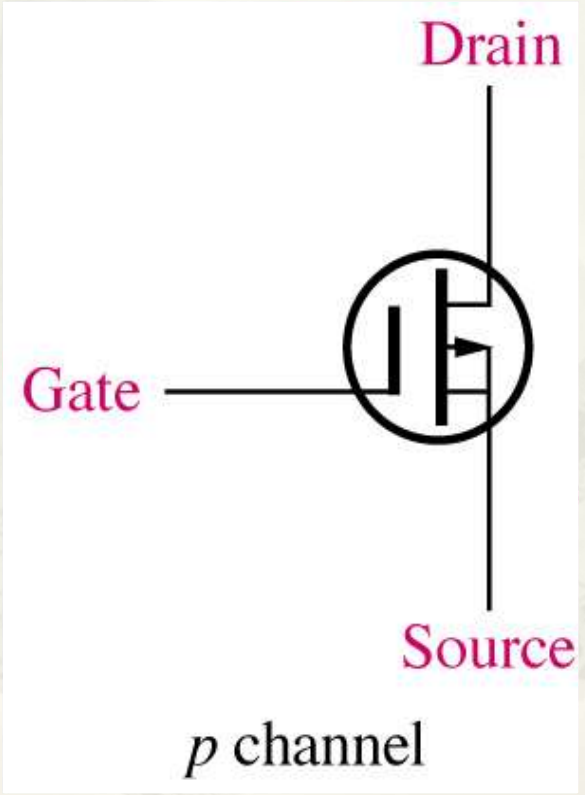
- **Two operation modes of D-MOSFET**
  - Depletion mode: V_{GS} negative and less than $V_{GS(off)}$
  - Enhancement mode: V_{GS}positive



Depletion MOSFET (D-MOSFET)

- **D-MOSFET Symbols**

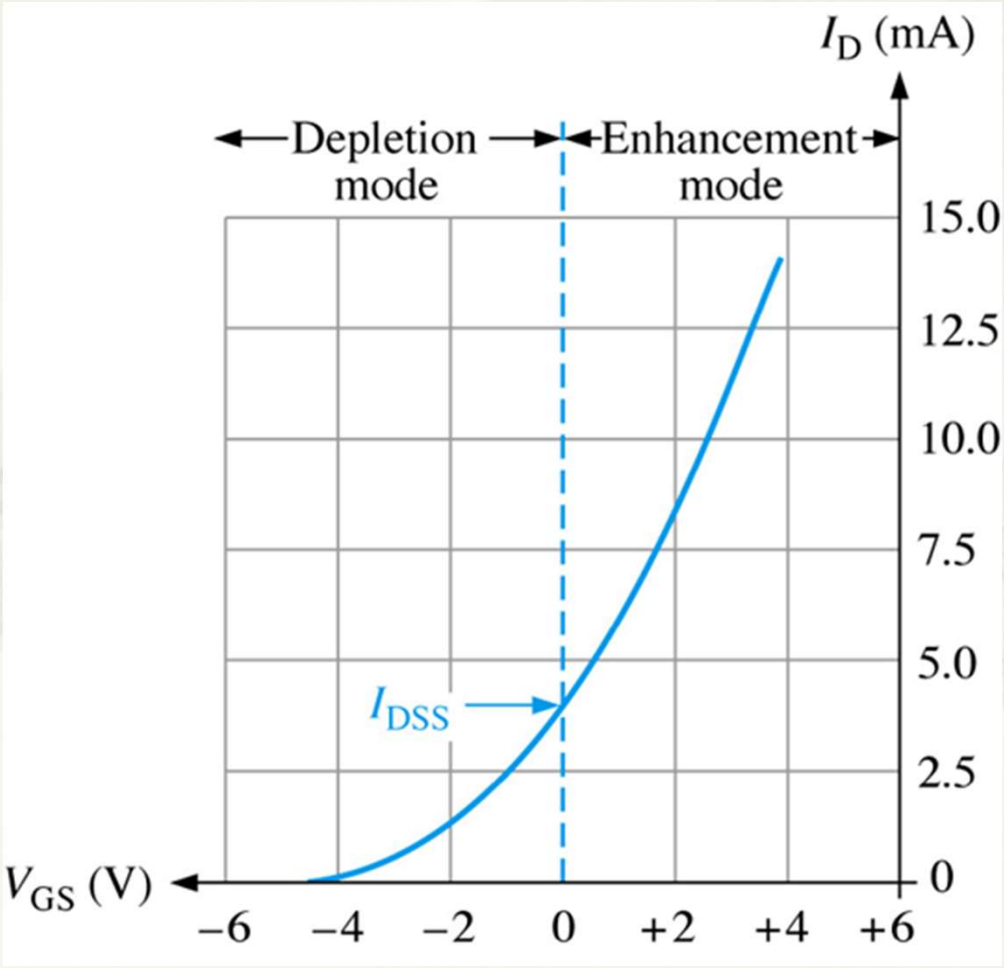




The substrate, indicated by the arrow, is normally (but not always) connected internally to the source.

Depletion MOSFET (D-MOSFET)

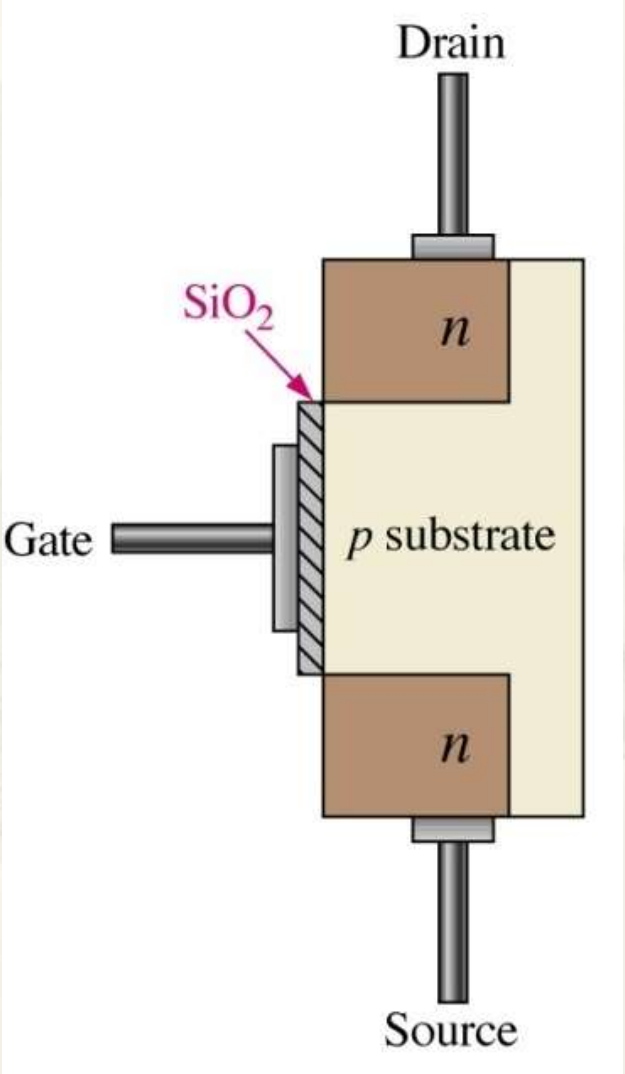
- **D-MOSFET transconductance curve**



Operation with currents higher than I_{DSS} is permissible with a D-MOSFET but not with a JFET.

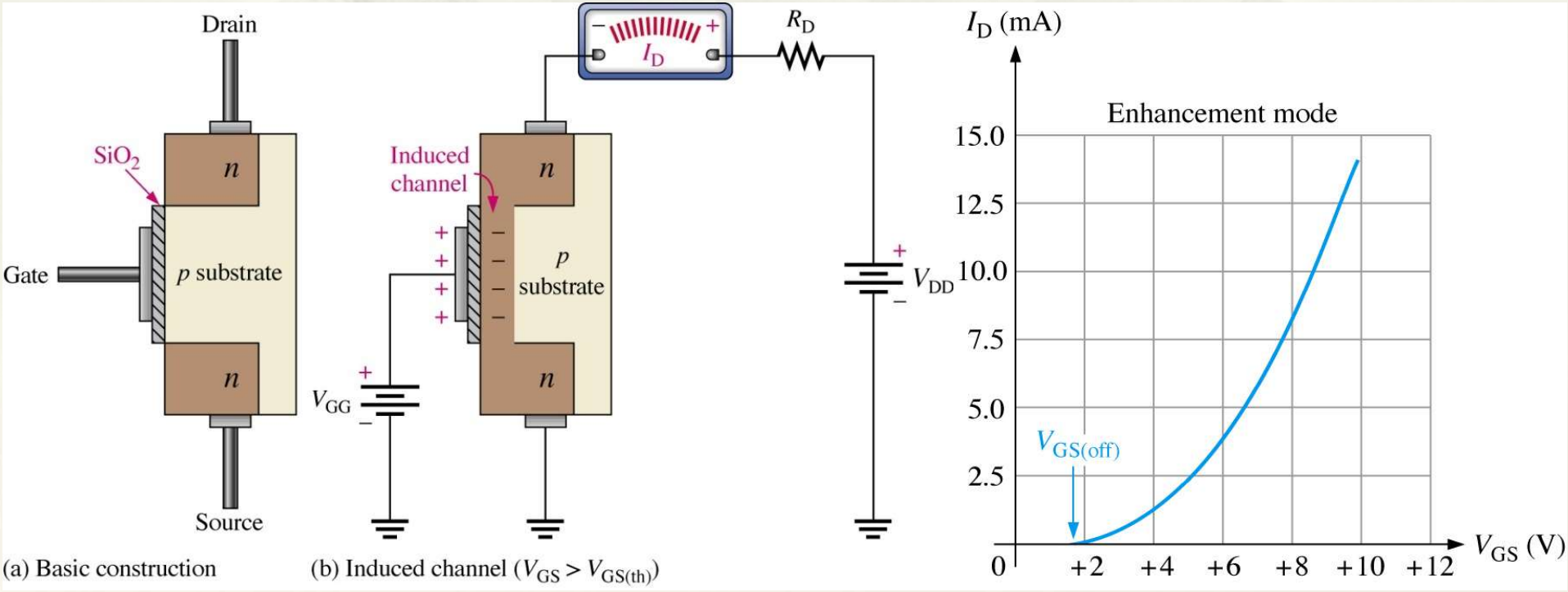
Enhancement MOSFET (E-MOSFET)

- **Enhancement MOSFET (E-MOSFET)**
  - That operates only in the enhancement mode and has no depletion mode.
  - The substrate extends completely to the SiO_2 layer.
- **Structure of E-MOSFET**



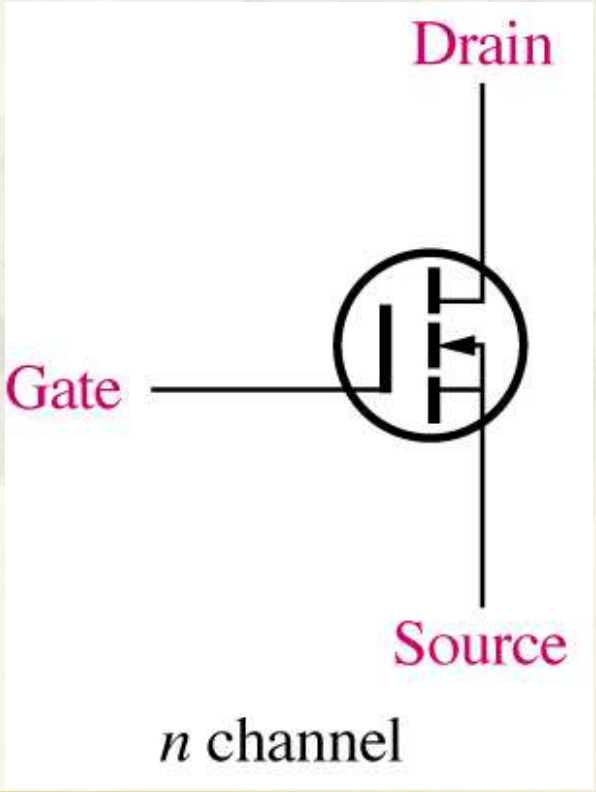
Enhancement MOSFET (E-MOSFET)

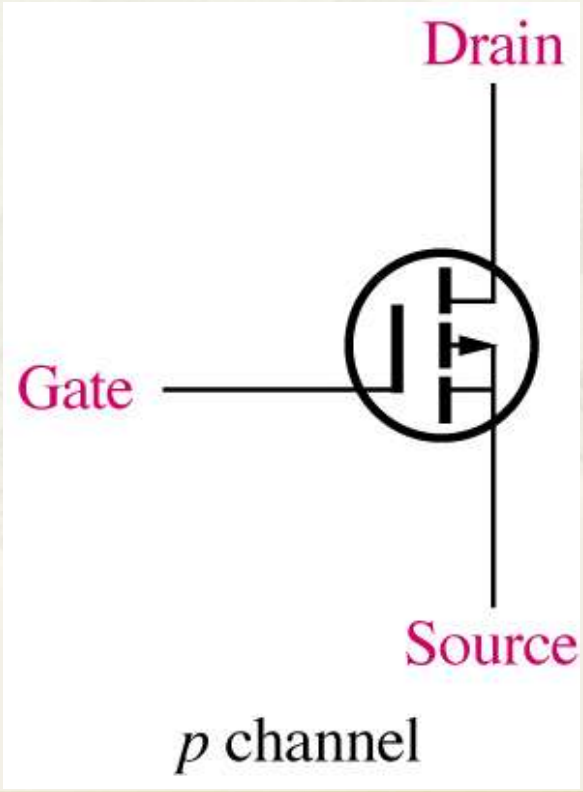
- **E-MOSFET construction and operation**
  - $V_{GS(th)}$: threshold voltage



Enhancement MOSFET (E-MOSFET)

- **E-MOSFET Symbols**
  - The broken line symbolize the absence of a physical channel.





FET Parameters

- **Static Electrical Characteristics**
  - Gate Threshold Voltage $(V_{GS(th)})$
  - Gate Cutoff Voltage $(V_{GS(off)})$
  - Zero Gate Voltage Drain Current (I_{DSS})
  - DC Input Resistance $(R_{GS(DC)})$
  - Gate-body Leakage Current (I_{GSS})
  - Static Drain-source On Resistance $(R_{DS(ON)})$

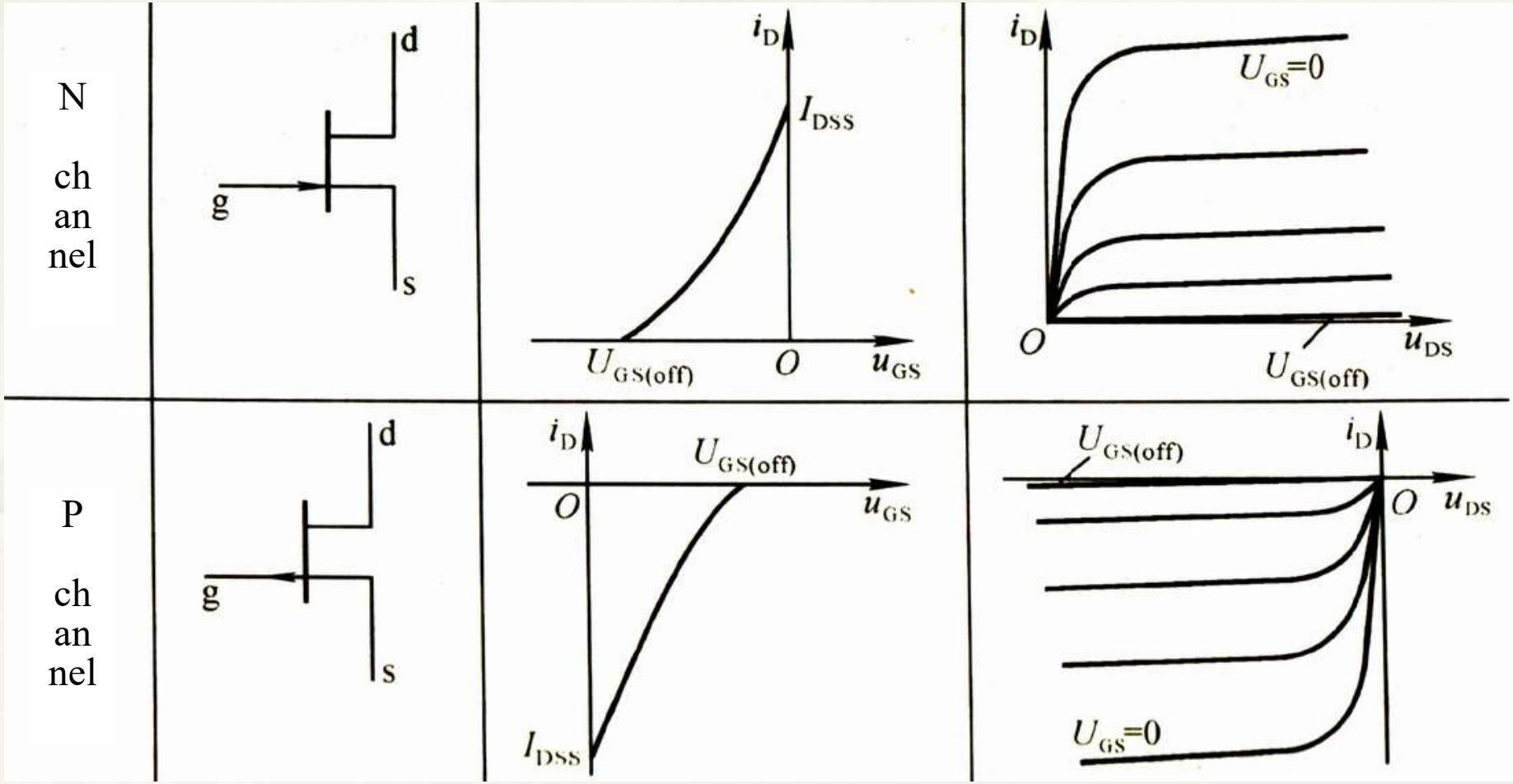
FET Parameters

- **Dynamic Electrical Characteristics**
  - Forward Transconductance (g_m)
  - Capacitance Between Leads (C_{gs}, C_{ds}, C_{gd})
- **Absolute Maximum Ratings**
  - Gate-source Breakdown Voltage $(V_{(BR)GS})$
  - Drain-source Breakdown Voltage $(V_{(BR)DS})$
  - Maximum Drain Current (I_{DM})
  - Maximum Power Dissipation (P_{DM})

Datasheet of an Actual FET

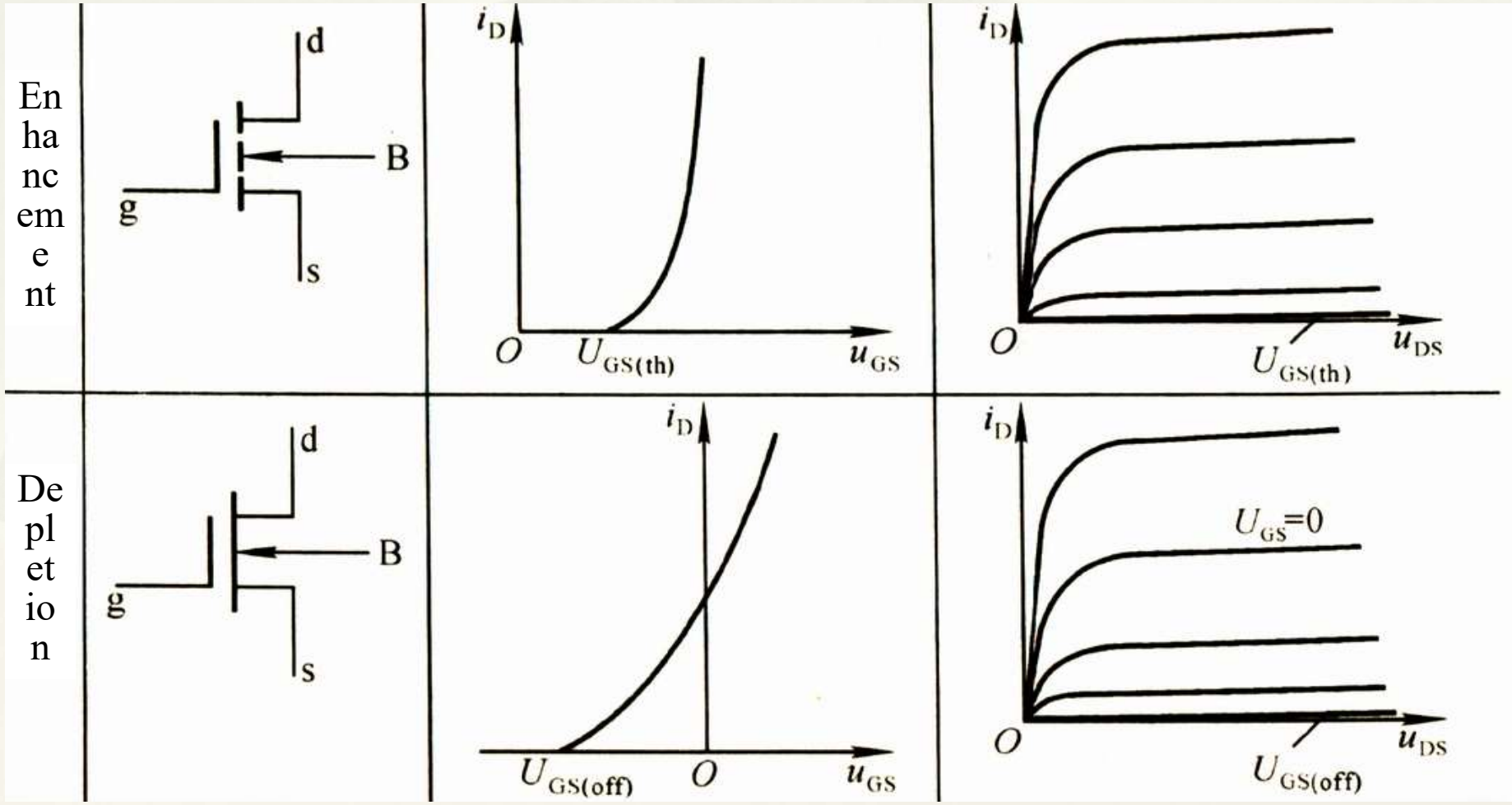
FET Characteristic Curve

- **JFET**



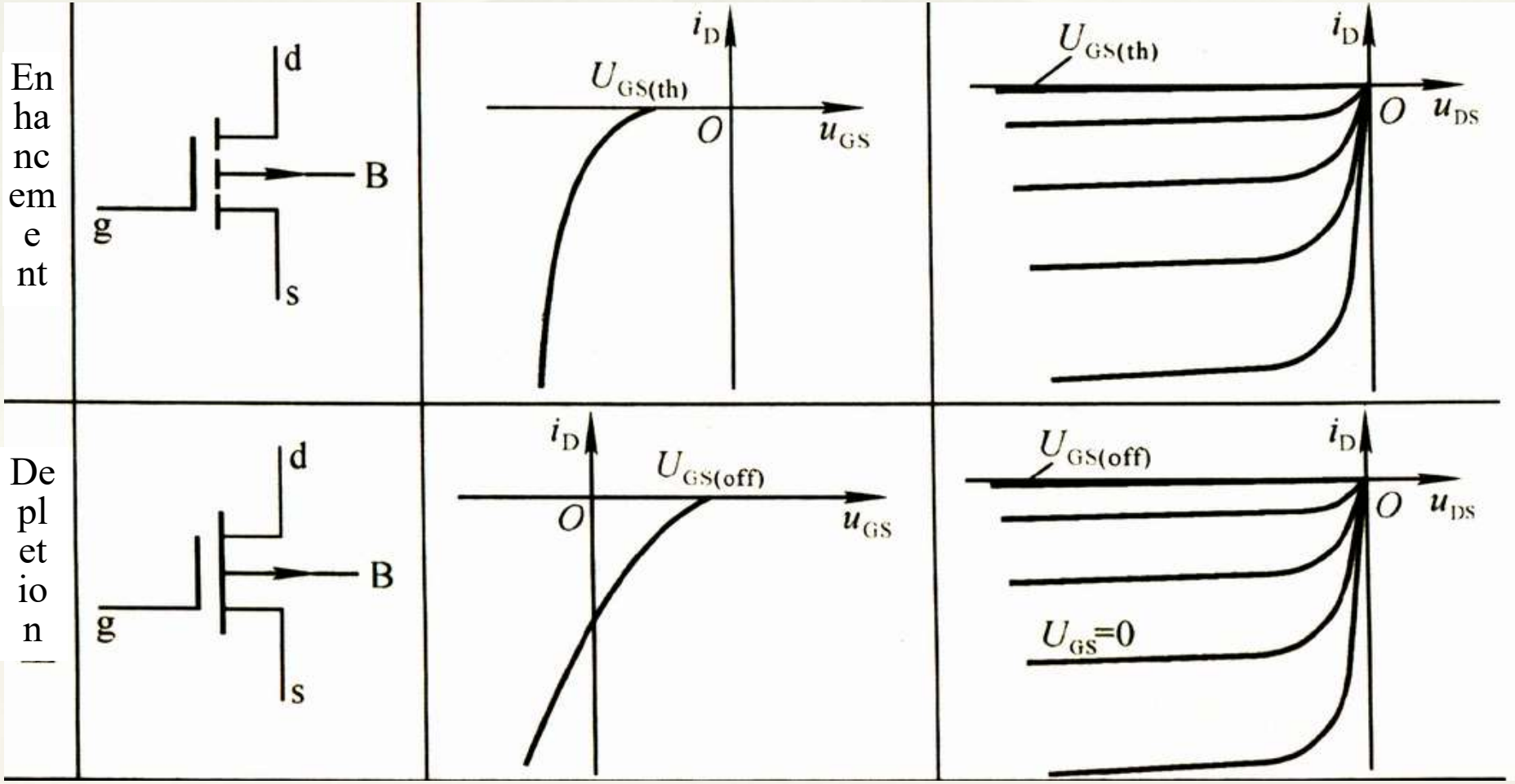
FET Characteristic Curve

- **N-Channel MOSFET**



FET Characteristic Curve

- **P-Channel MOSFET**



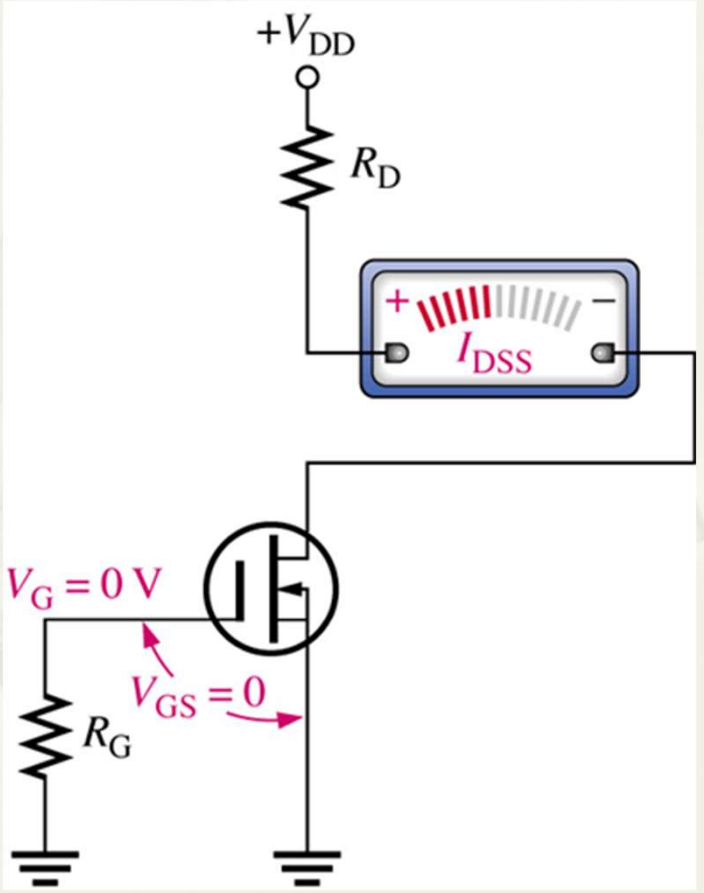
4-5 MOSFET Biasing

- D-MOSFET Bias
- E-MOSFET Bias

Bias establishes the appropriate dc operating conditions that provide a stable operating point for centering the ac signal.

D-MOSFET Bias

- **Operating with the gate-to-source voltage (V_{GS})**
  - Negative: depletion mode
  - Positive: enhancement mode
- **Zero bias**
  - The most basic bias method
  - Between depletion and enhancement
  - $V_{GS} = 0\text{V}, I_D = I_{DSS}$
  - $V_D = V_{DD} - I_{DSS}R_D$



Example 4-5

Problem

Determine the drain-to-source voltage in the circuit of Figure 4–23. The MOSFET data sheet gives $I_{DSS} = 12$ mA.

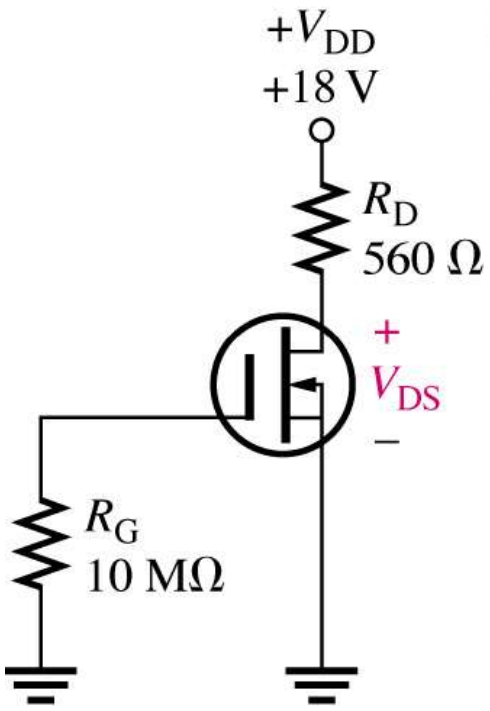


Figure 4-23

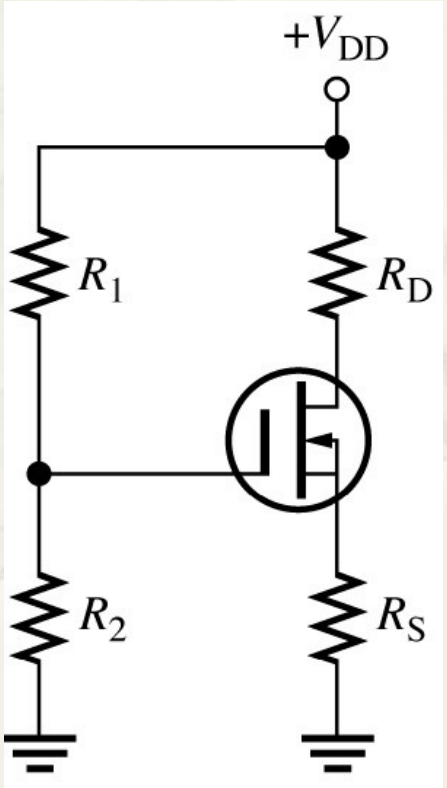
Solution

Since $I_D = I_{DSS} = 12$ mA, the drain-to-source voltage is

$$V_{DS} = V_{DD} - I_{DSS}R_D = 18\text{ V} - (12\text{ mA})(560\ \Omega) = \mathbf{11.28\ V}$$

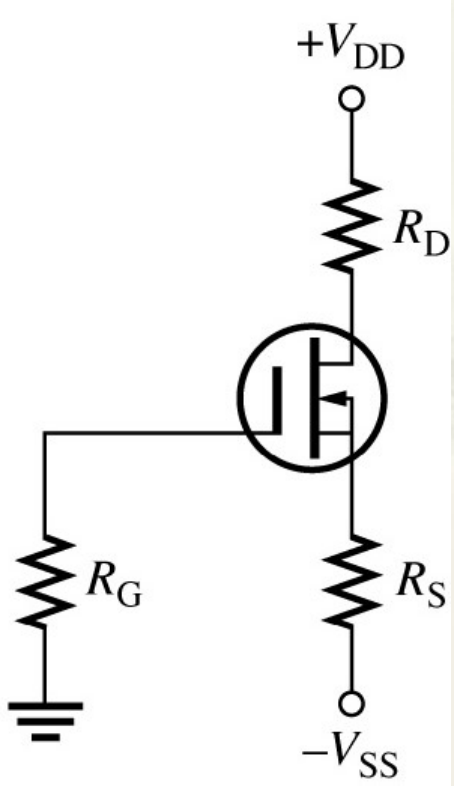
D-MOSFET Bias

- **Voltage-divider and self-bias**
  - The voltage at the gate is quite accurate for any FET device because if the negligible loading effect.
  - $V_G = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$



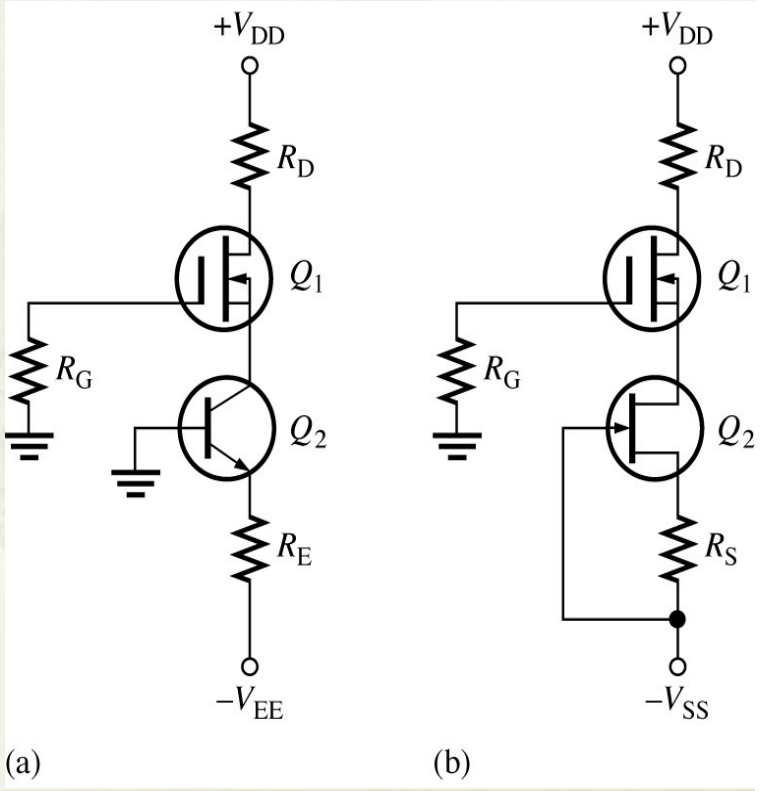
D-MOSFET Bias

- **Voltage-divider and self-bias**
  - The voltage at the gate is quite accurate for any FET device because if the negligible loading effect.
  - $V_G = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$
- **Source bias**
  - For double power supplies
  - $V_G = 0V$



D-MOSFET Bias

- **Voltage-divider and self-bias**
  - The voltage at the gate is quite accurate for any FET device because if the negligible loading effect.
  - $V_G = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$
- **Source bias**
  - For double power supplies
  - $V_G = 0V$
- **Current source bias**
  - Used in operational amplifiers



Example 4-6

Problem

What is the drain current for the D-MOSFET in Figure 4–26?

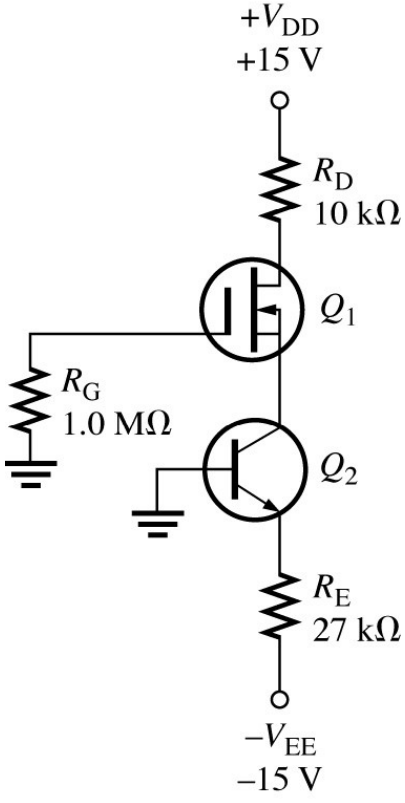


Figure 4-26

Solution

The base of Q_2 is connected to ground; therefore, the emitter is −0.7 V. The voltage across R_E is

$$V_{R_E} = -0.7\text{ V} - (-15\text{ V}) = 14.3\text{ V}$$

The current in the emitter resistor is

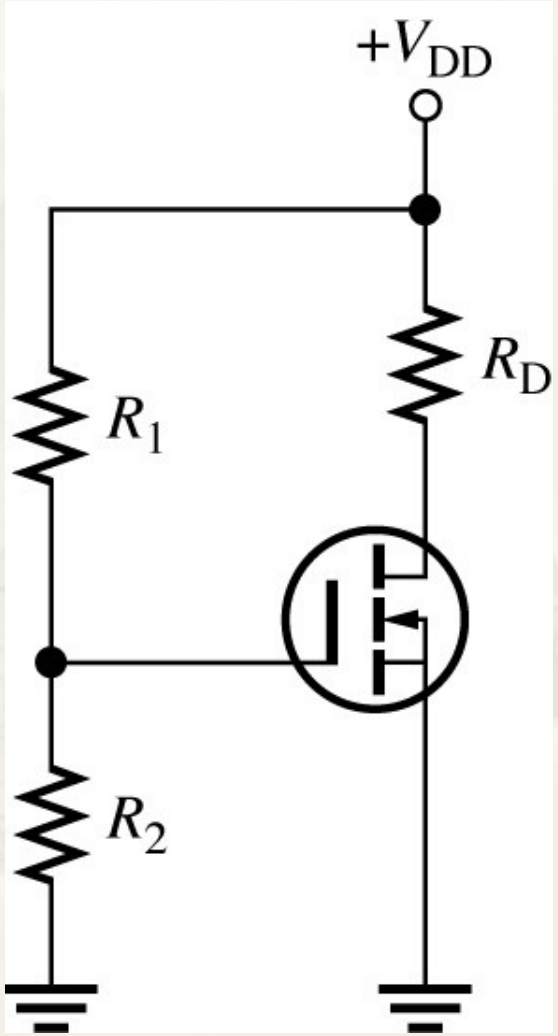
$$I_{R_E} = \frac{14.3\text{ V}}{27\text{ k}\Omega} = 0.53\text{ mA}$$

The D-MOSFET acts as a load for the BJT. Consequently, the drain current is approximately equal to the emitter current.

$$I_D = I_{R_E} = \mathbf{0.53\text{ mA}}$$

E-MOSFET Bias

- **Voltage-divider and self-bias**
  - With the voltage-divider bias arrangement, the gate voltage will be more positive than the source by an amount exceeding $V_{GS(th)}$
  - $V_G = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$



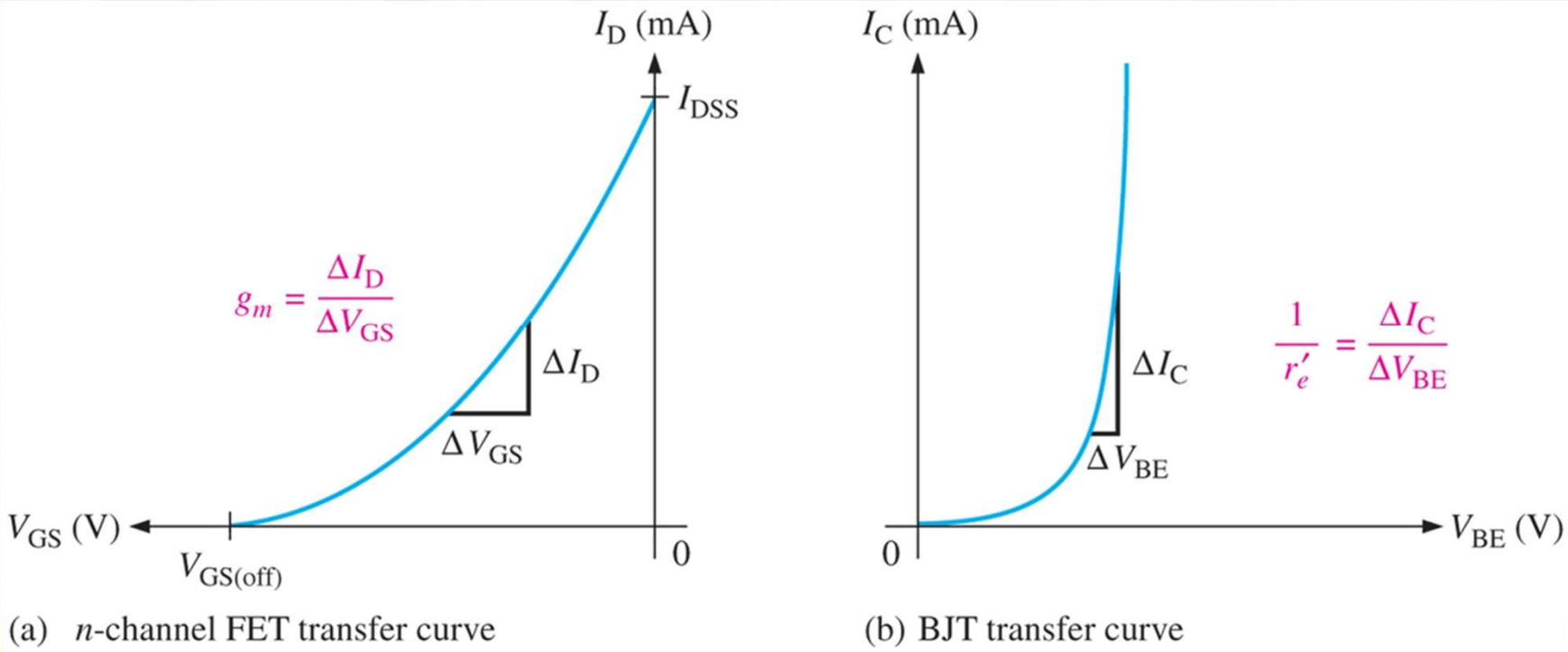
4-6 FET Linear Amplifiers

- Transconductance of FETs
- Common-Source (CS) Amplifiers
- Common-Drain (CD) Amplifiers

Transconductance of FETs

- The output drain current is controlled by the input gate voltage.
- Transconductance of FETs: $g_m = \frac{I_d}{V_{gs}}$
- The transconductance depends on the drain current.

Comparison of the transfer curve for an n-channel FET with a BJT

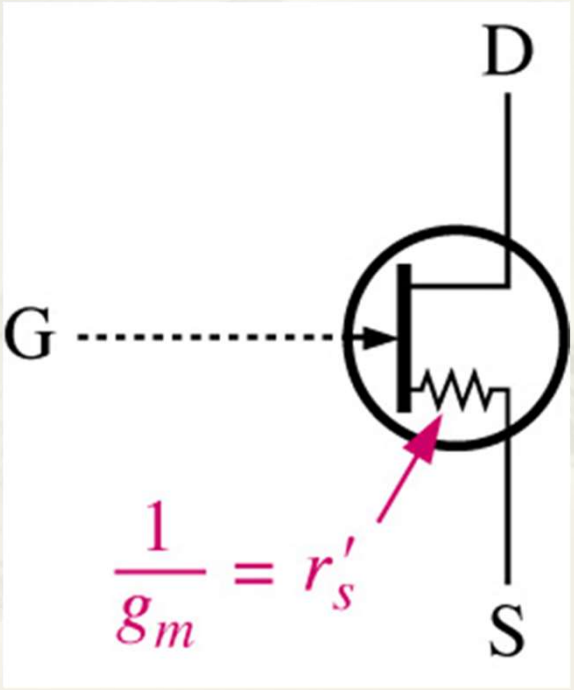


(a) *n*-channel FET transfer curve

(b) BJT transfer curve

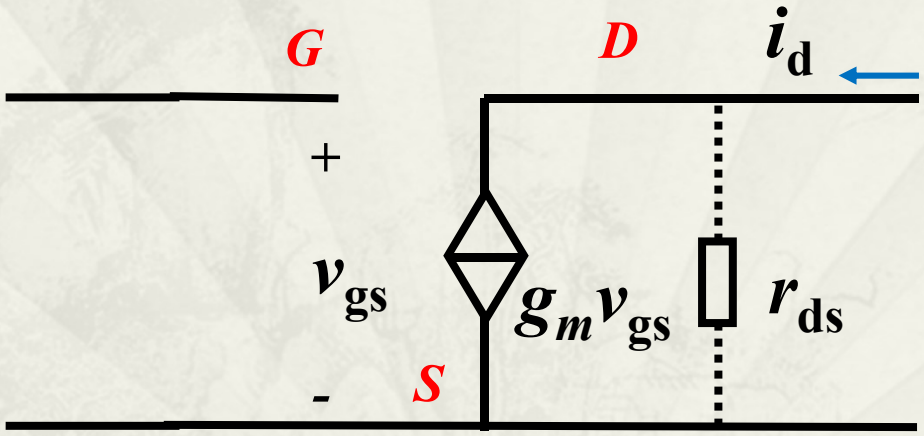
Transconductance of FETs

- **AC source resistance of the FET(r'_s)**
  - $r'_s = \frac{1}{g_m}$
  - Analogous to r'_e for BJTs
  - The dotted line is a reminder that gate current is negligible
  - Not as predictable as r'_e
  - Generally larger than r'_e



Transconductance of FETs

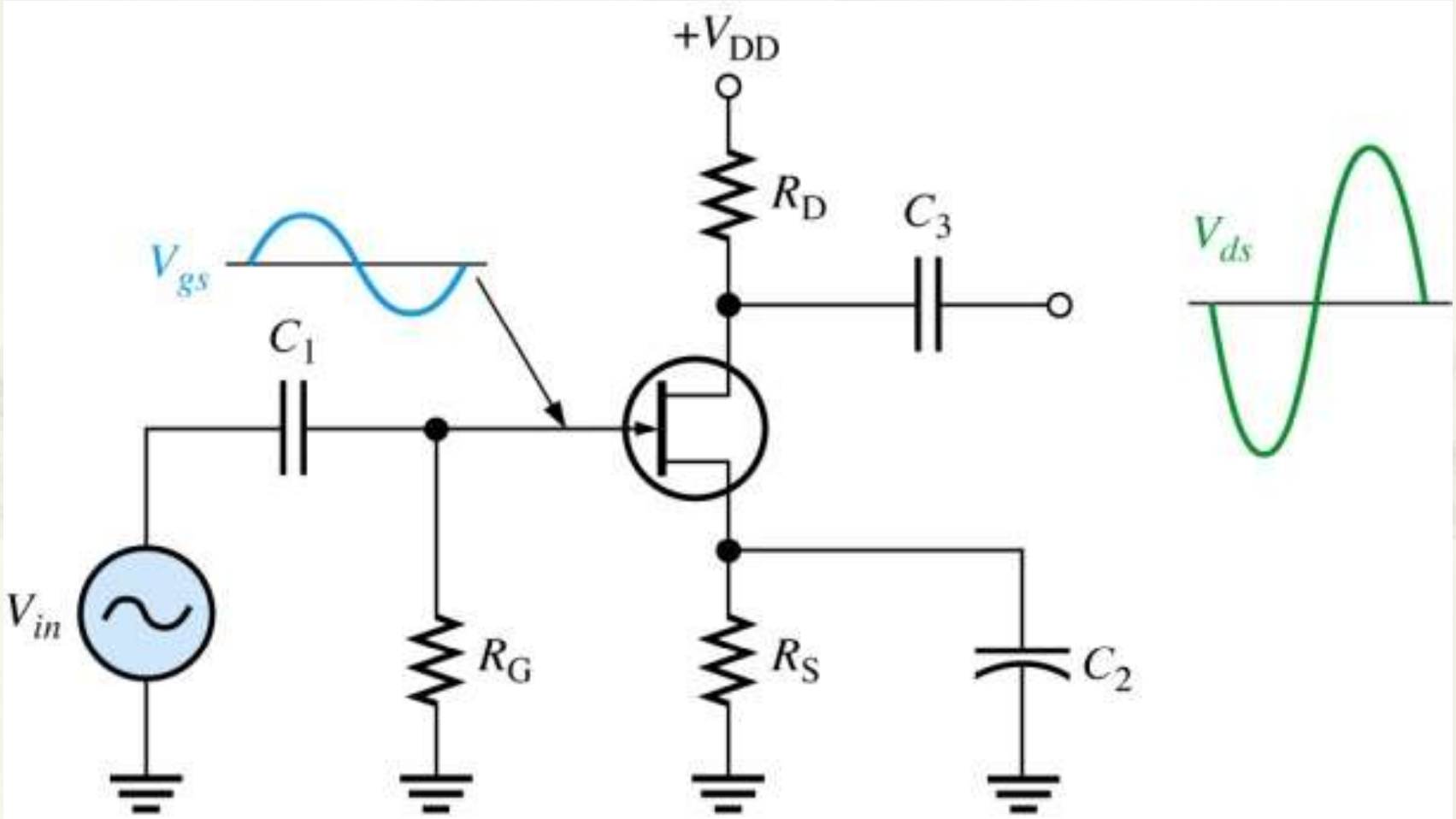
- **AC model of a FET**
  - available for low-frequency small signals



Note: r_{ds} is very high, can be neglected

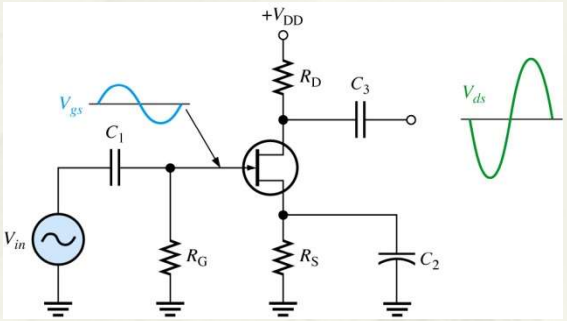
Common-Source Amplifiers

- **JFET Common-Source (CS) Amplifier**



Common-Source Amplifiers

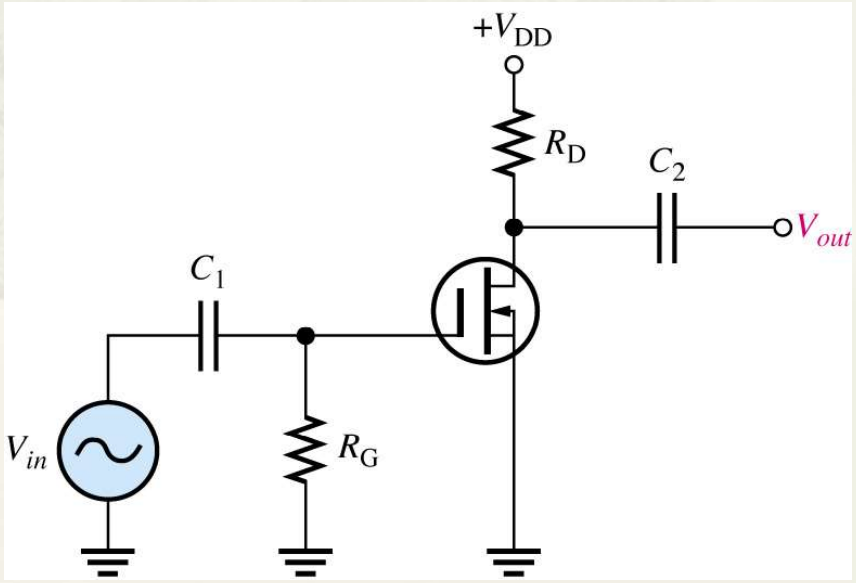
- **JFET Common-Source (CS) Amplifier**
  - Self-biased: created by the drop across R_S
  - The source is capacitively coupled to the gate by C_1
  - R_G serves two purpose:
    - ✓ Keeps the gate at approximately 0V dc
    - ✓ Prevents loading of the ac signal source



  - The bypass capacitor C_2
    - ✓ Keeps the source of the FET effectively at ac ground
  - The drain-to source voltage is 180° out of phase with the gate-to-source voltage

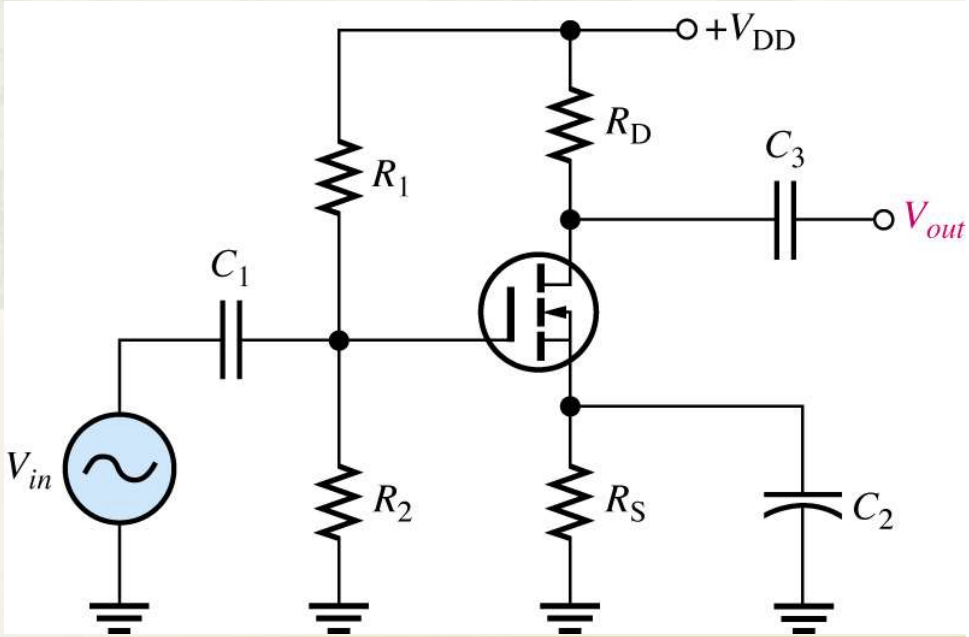
Common-Source Amplifiers

- **D-MOSFET Common-Source (CS) Amplifier**
  - Zero-biased: created by $R_{\mathbf{G}}$, $R_{\mathrm{G}} = 0\mathrm{V}$
  - The source is capacitively coupled to the gate by C_1
  - The negative swing in V_{gs} produces the depletion mode, and I_d decreases.
  - The positive swing in V_{gs} produces the enhancement mode, and I_d increases.



Common-Source Amplifiers

- **E-MOSFET Common-Source (CS) Amplifier**
  - Voltage-divider biased: $V_{GS} > V_{GS(off)}$
  - The source is capacitively coupled to the gate by C_1
  - The signal voltage produces a swing in V_{gs} above and below its Q-point value.
  - Operation entirely in the enhancement mode.



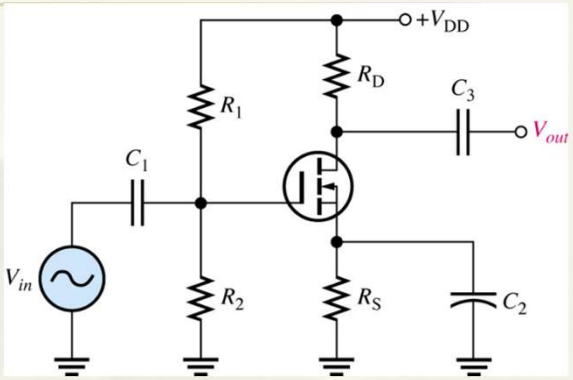
Common-Source Amplifiers

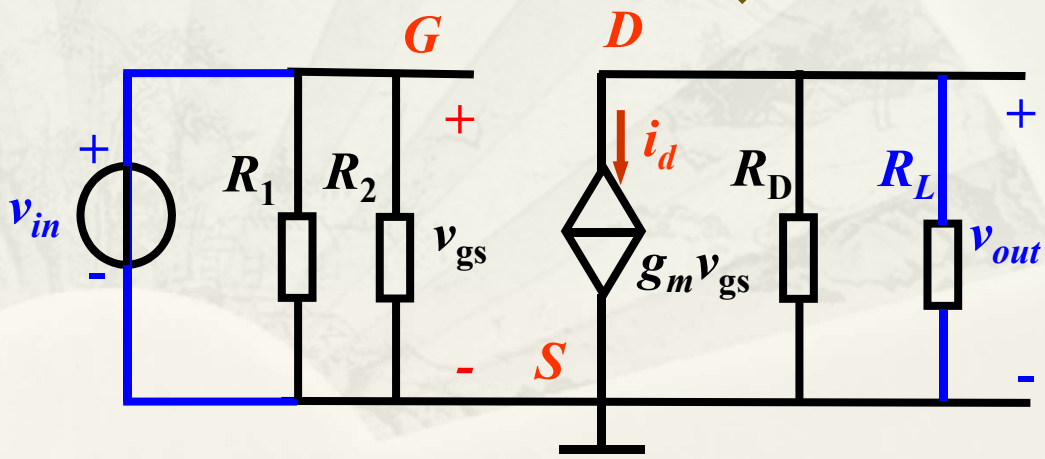
- **Voltage Gain (A_v)**
  - $A_v = \dfrac{V_{out}}{V_{in}}$

$$= -\frac{I_d R_d}{V_{gs}} \qquad (R_d = R_D // R_L)$$

$$= -g_m R_d$$

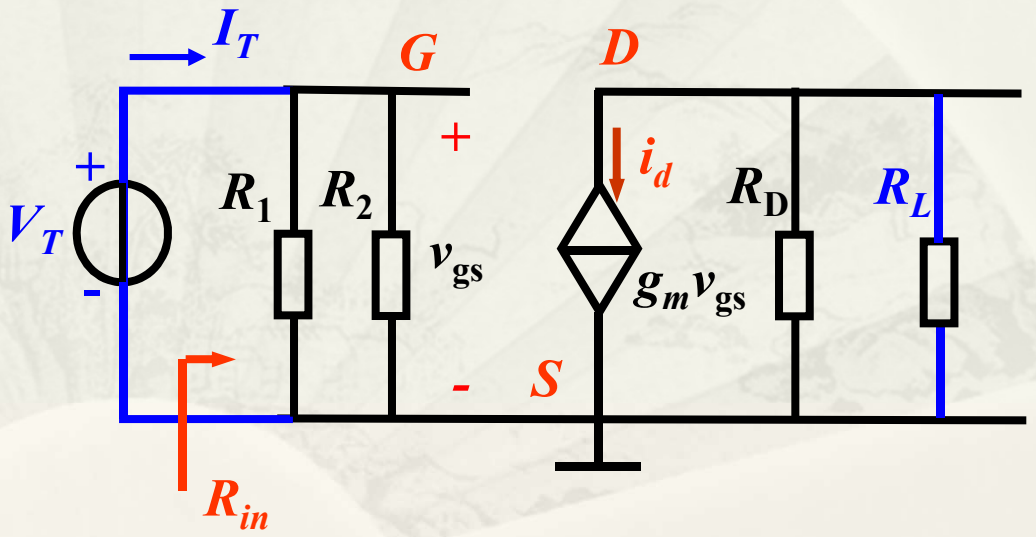
Since, $(g_m = I_d / V_{gs})$





Common-Source Amplifiers

- **Input Resistance (R_{in})**
  - The transistor's internal resistance is ignored, the input resistance seen by the signal source is determined only by the bias resistors.
  - $R_{in} \cong R_1 \,||\, R_2$

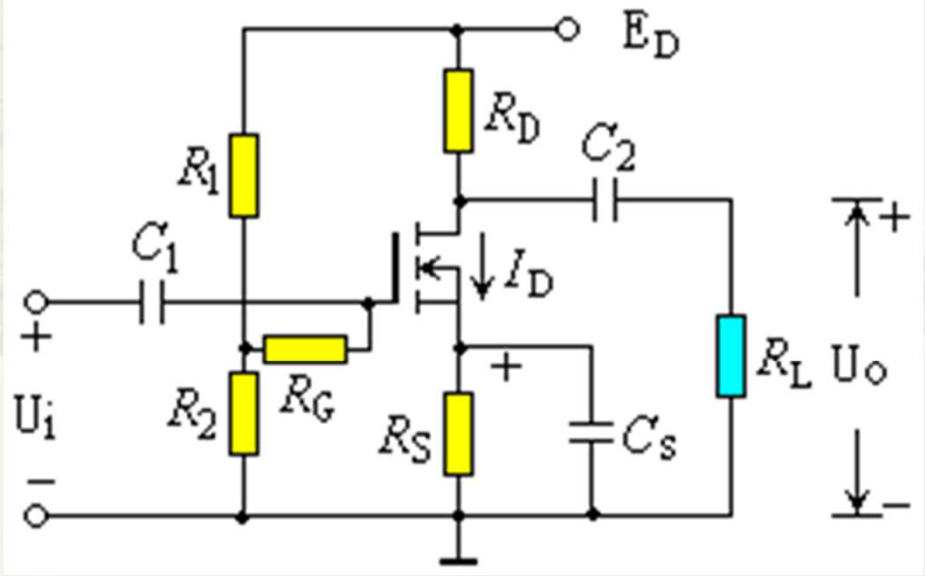


Common-Source Amplifiers

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  - The transistor's internal resistance is ignored, the input resistance seen by the signal source is determined only by the bias resistors.

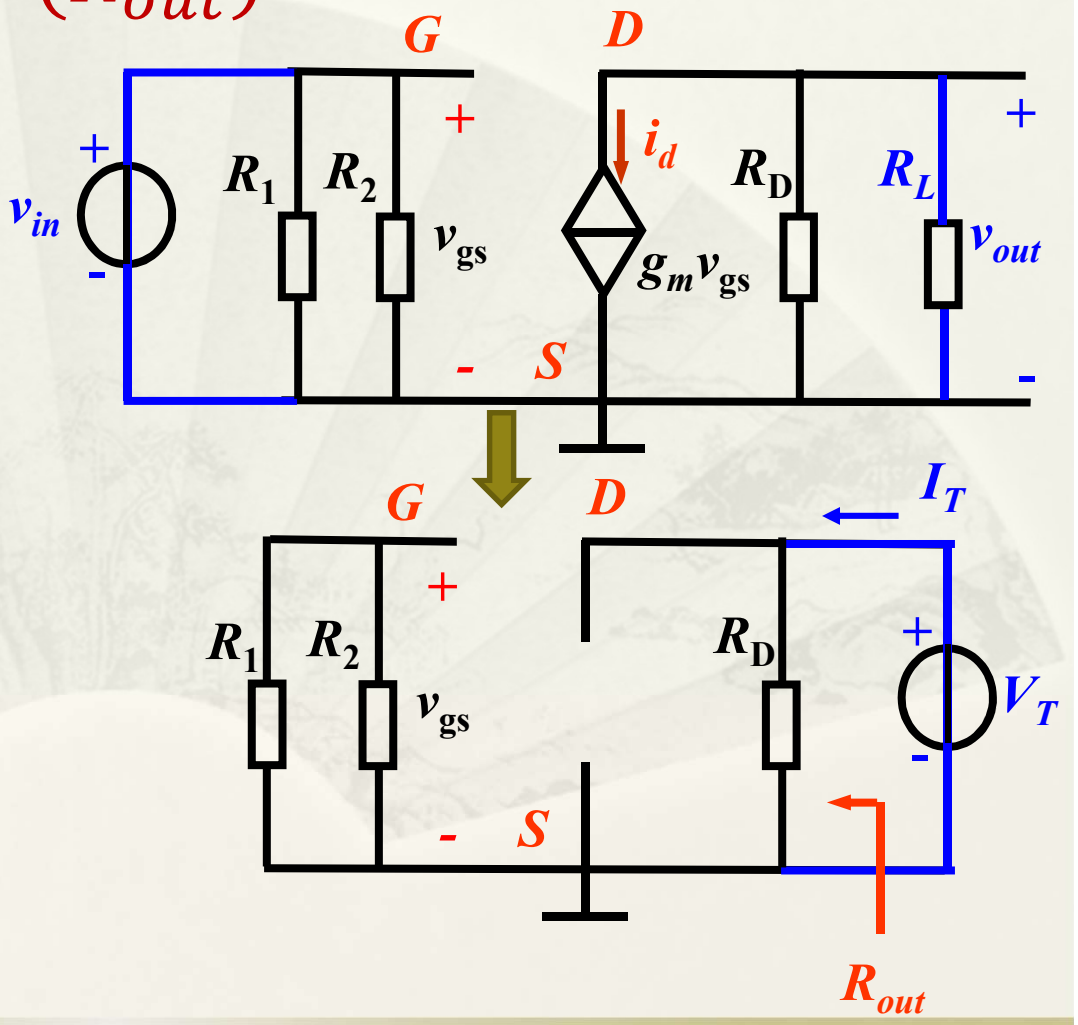
$$R_{in} \cong R_1 \,||\, R_2 + R_G$$

What about the output resistance?



Common-Source Amplifiers

- **Output Resistance (R_{out})**
  - $R_{out} \cong R_{\mathrm{D}}$

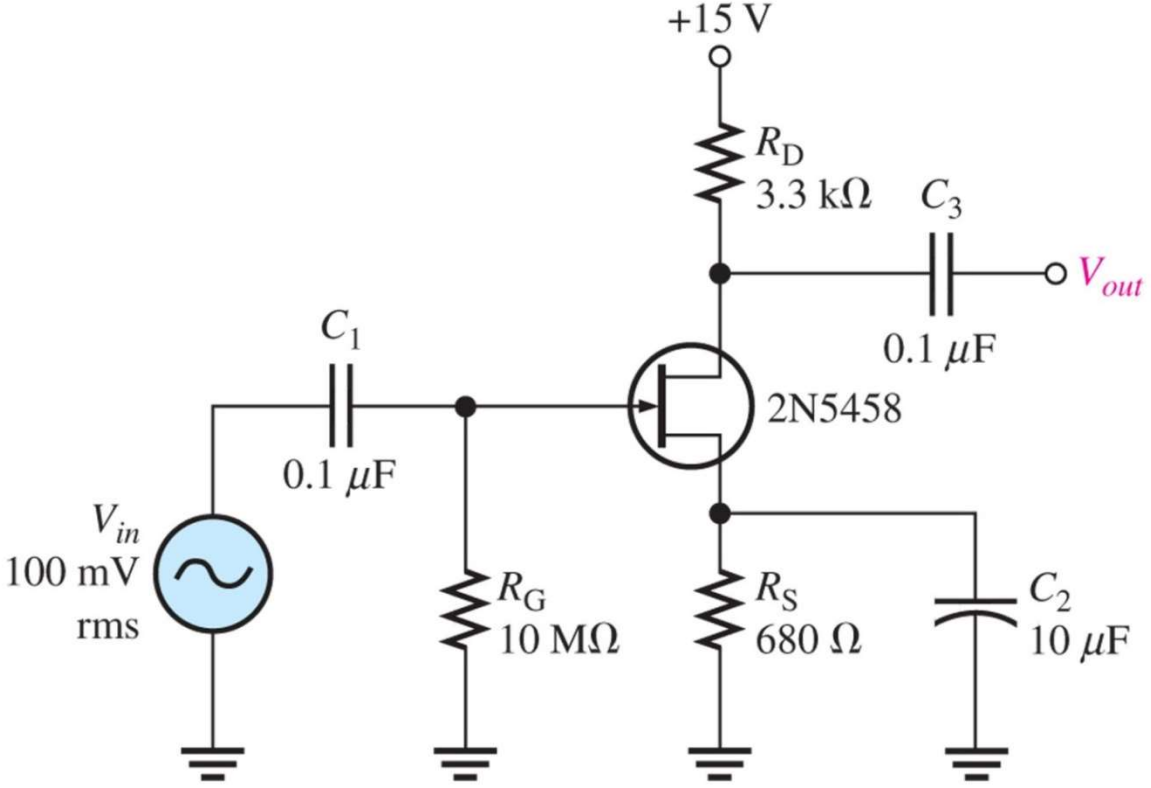


Example 4-7

Problem

(a) What is the total output voltage (dc + ac) of the amplifier in Figure 4–33? The g_m is 2500 μS and I_D is 1.7 mA.

(b) What is the input resistance seen by the signal source?



Example 4-7

Solution

(a) First, find the dc output voltage.

$$V_D = V_{DD} - I_D R_D = 15\ \text{V} - (1.7\ \text{mA})(3.3\ \text{k}\Omega) = 5.6\ \text{V}$$

Next, find the voltage gain.

$$A_v = -g_m R_d = -(2500\ \mu\text{S})(3.3\ \text{k}\Omega) = -8.25$$

Alternatively, the voltage gain could be found by computing r'_s and using the ratio of ac drain resistance to ac source resistance.

$$r'_s = \frac{1}{g_m} = \frac{1}{2500\ \mu\text{S}} = 400\ \Omega$$

Example 4-7

$$A_v = -\frac{R_d}{r_s'} = -\frac{3.3\ \text{k}\Omega}{400\ \Omega} = -8.25$$

The output voltage is the gain times the input voltage.

$$V_{out} = A_v V_{in} = (-8.25)(100\ \text{mV}) = -825\ \text{mV rms}$$

The negative sign indicates the output waveform is inverted.

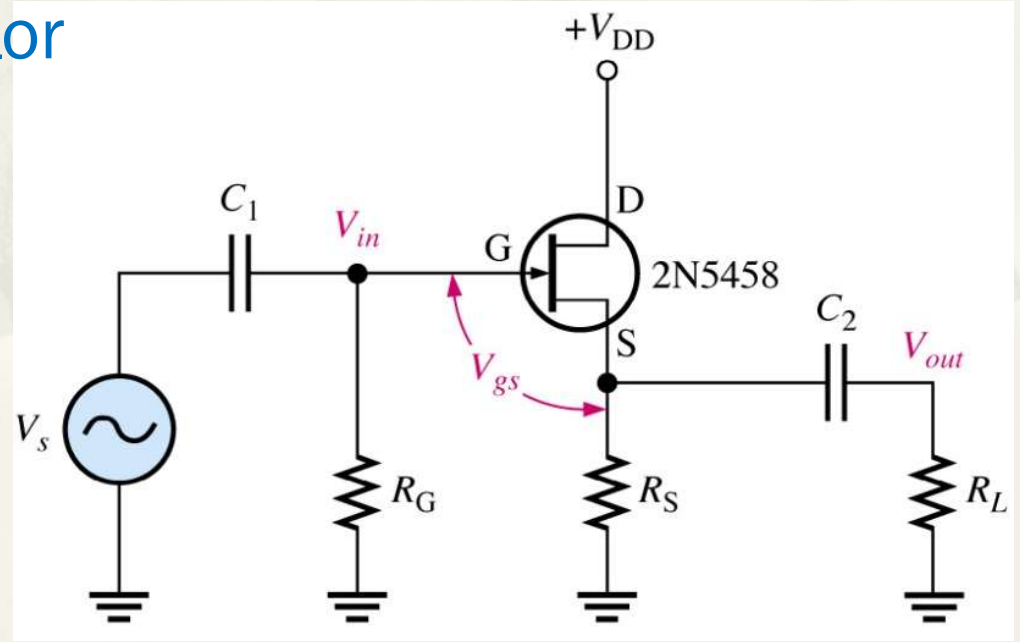
The total output voltage is an ac signal with a peak-to-peak value of 0.825 V × 2.828 = **2.33 V,** riding on a dc level of **5.6 V.**

(b) The input resistance is

$$R_{in} \cong R_G = \mathbf{10\ M\Omega}$$

Common-Drain (CD) Amplifier

- **Common-Drain (CD) Amplifier (source-follower)**
  - Self-biased
  - The input signal is capacitively coupled to the gate
  - The output signal is capacitively coupled to the load
  - There is no drain resistor

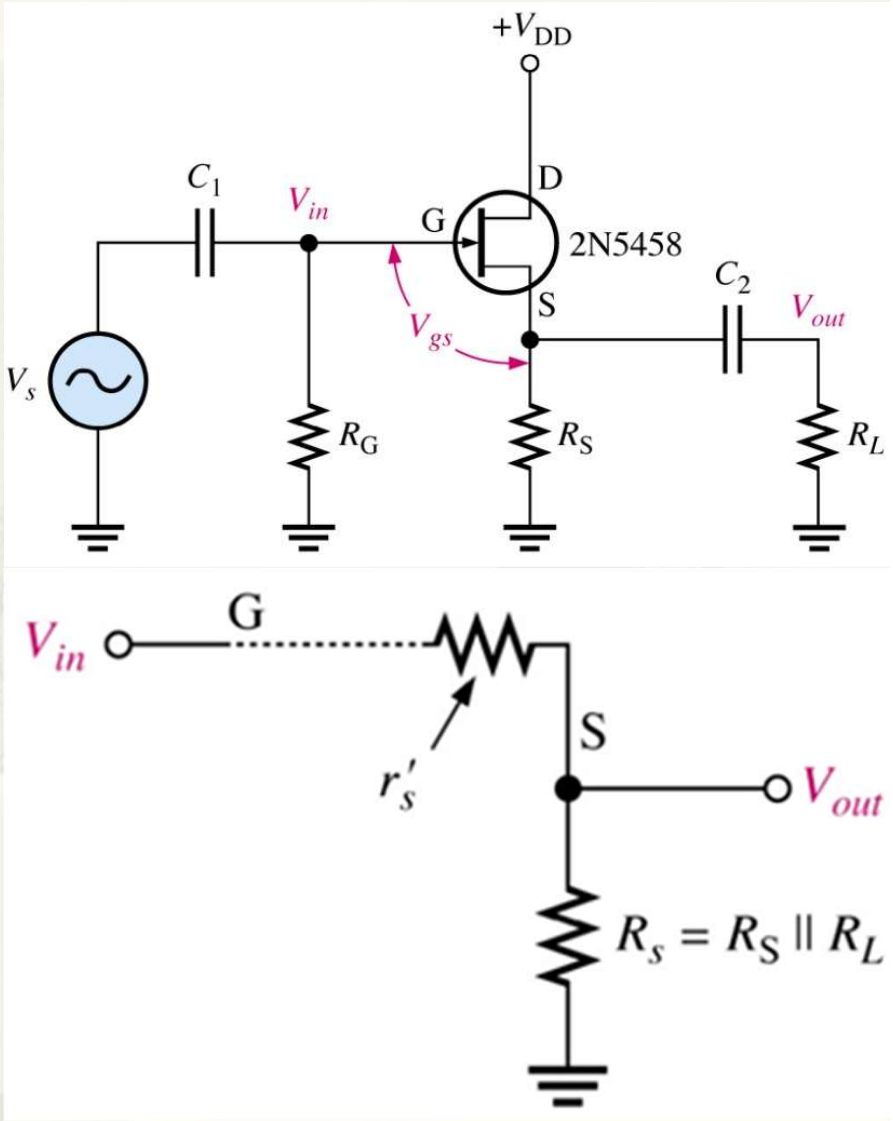


Common-Drain (CD) Amplifier

- **Voltage Gain**
  - $V_{out} = V_{in}\left(\frac{R_s}{r_s' + R_s}\right)$
  - $A_v = \frac{R_s}{r_s' + R_s} \qquad \left(r'_s = \frac{1}{g_m}\right)$

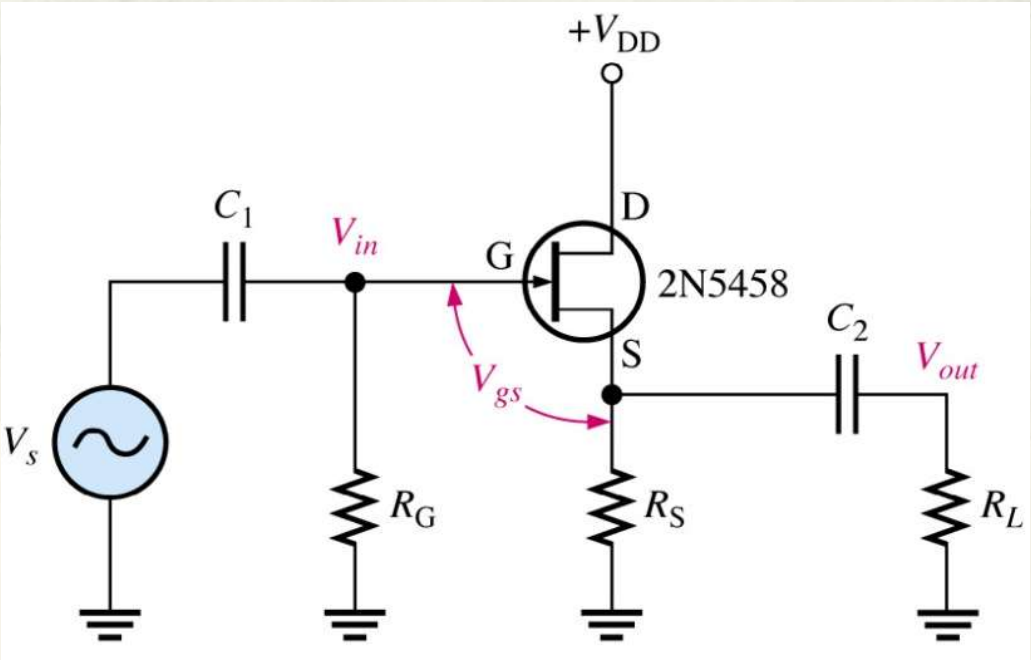
$$= \frac{g_m R_s}{1 + g_m R_s}$$

  - Ideal voltage gain: 1
  - In practice, less, typically 0.5 – 1.0
  - In phase with the input



Common-Drain (CD) Amplifier

- **Input Resistance**
  - Ignore the very high resistance of the transistor's input
  - For self-bias, $R_{in} \cong R_G$
  - For voltage-divider bias, $R_{in} \cong R_1 \,||\, R_2$



Example 4-8

Problem

The *p*-channel JFET shown in Figure 4–35 has a transconductance between 2000 μS and 6000 μS. Determine the minimum and maximum voltage gain.

Solution

The maximum value of r'_s is

$$r'_s = \frac{1}{g_m} = \frac{1}{2000\ \mu\text{S}} = 500\ \Omega$$

The ac source resistance, R_s, is simply the load resistor, R_L, which is R_S. Substituting into Equation 4–9, the minimum voltage gain is

$$A_{v(\text{min})} = \frac{R_s}{r'_s + R_s} = \frac{10\ \text{k}\Omega}{500\ \Omega + 10\ \text{k}\Omega} = \mathbf{0.95}$$

Example 4-8

The minimum value of r'_s is

$$r'_s = \frac{1}{g_m} = \frac{1}{6000\ \mu S} = 167\ \Omega$$

The maximum voltage gain is then

$$A_{v(\max)} = \frac{R_s}{r'_s + R_s} = \frac{10\ k\Omega}{167\ \Omega + 10\ k\Omega} = \mathbf{0.98}$$

Notice that the gain is slightly less than 1. When r'_s is small compared to the ac source resistance, then a good approximation is $A_v = 1$. Since the output voltage is at the source, it is in phase with the gate (input) voltage.

4-7 FET Switching Circuits

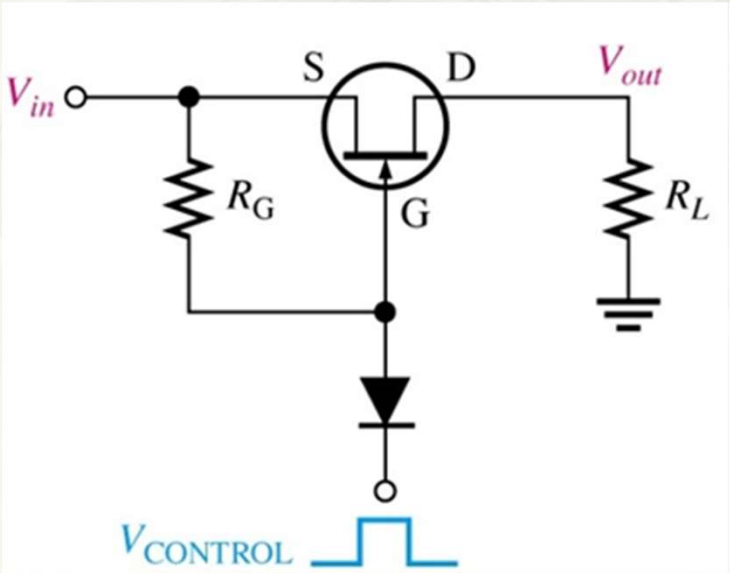
- Types of Solid-State Switches
- The Analog Switch
- The Digital Switch

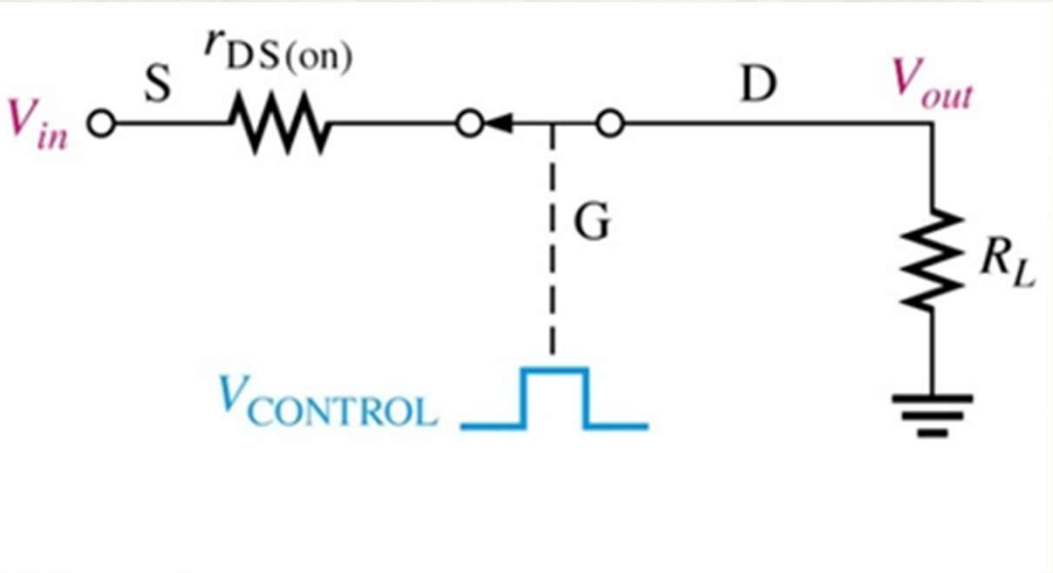
Types of Solid-State Switches

- **Analog Switches (ac switch)**
  - Takes advantages of the FET's low internal drain-source resistance when it is on and its high drain-source resistance when it is off
  - Connected in series with a signal
    - When closed: the signal should be passed to the load without regard to the waveform polarity and with no significant change.
    - When open: the signal should be completely blocked and the switch should appear as an open circuit.
- **Digital Switches (dc or logic switch)**
  - Frequently designed for power switching where current and voltages maybe large

The Analog Switch

- **JFET Analog Switch**
  - Potential problem: forward-biased pn junction
  - Resistor and diode
  - The on-state channel resistance ($r_{DS(on)}$)
  - $r_{DS(on)} = -\frac{V_{GS(off)}}{2I_{DSS}}$





Example 4-8

Problem

Compute $r_{DS(on)}$ for a JFET with the transconductance curve shown in Figure 4–37.

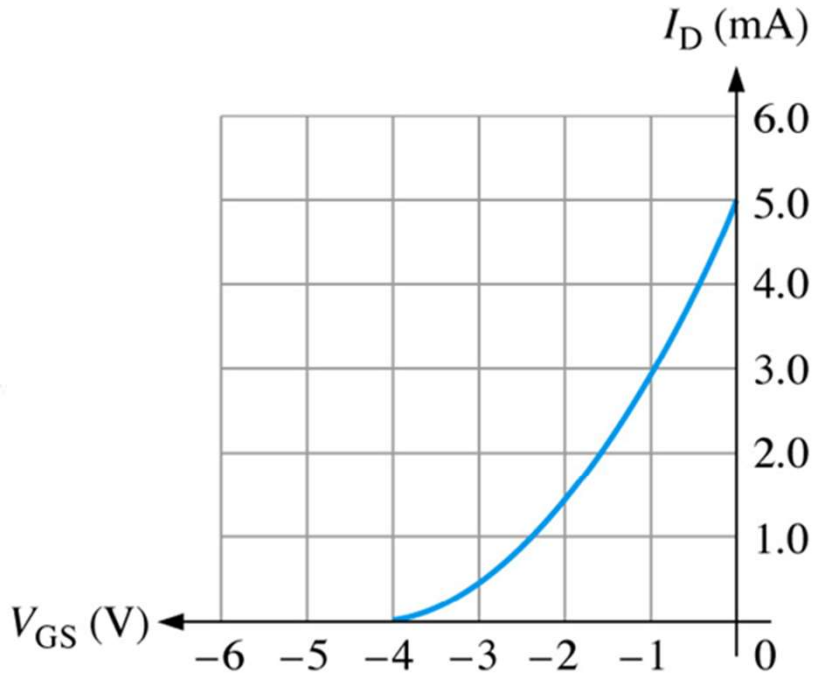


FIGURE 4–37

Example 4-8

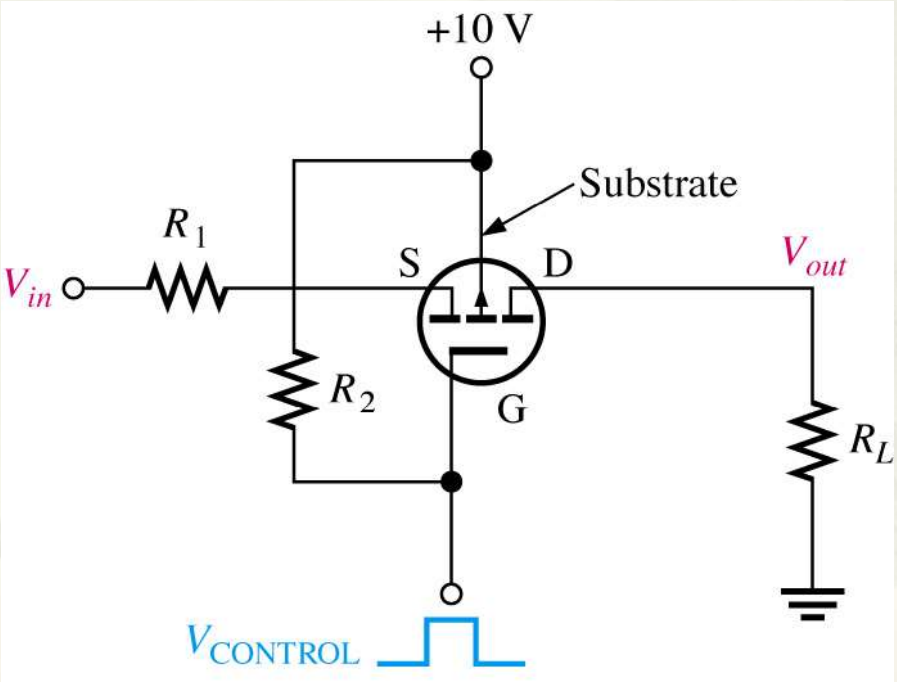
Solution

Reading from the graph, $V_{GS(off)} = -4$ V and $I_{DSS} = 5.0$ mA. Substituting into the equation for $r_{DS(on)}$ gives

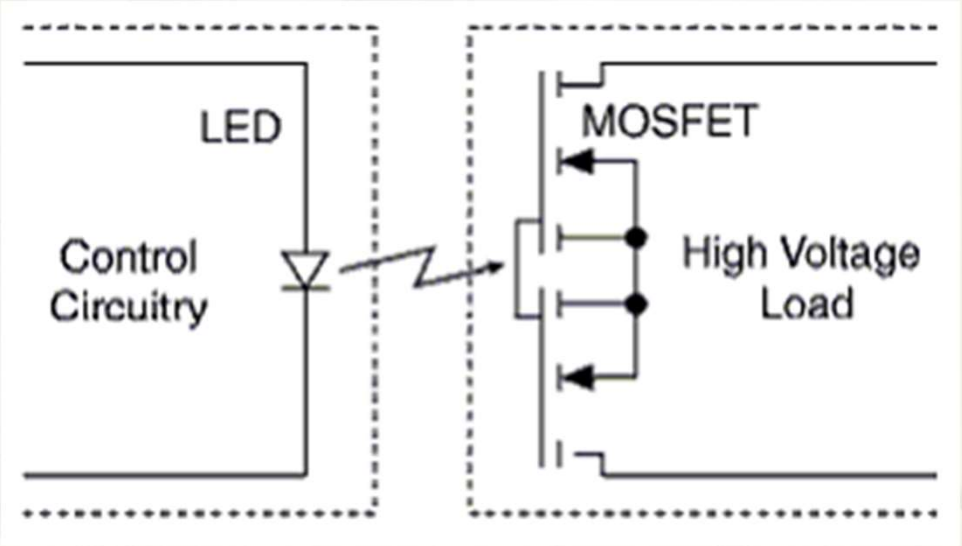
$$r_{DS(on)} = -\frac{V_{GS(off)}}{2I_{DSS}} = -\frac{-4\text{ V}}{2(5.0\text{ mA})} = \mathbf{400\ \Omega}$$

The Analog Switch

MOSFET Analog Switch

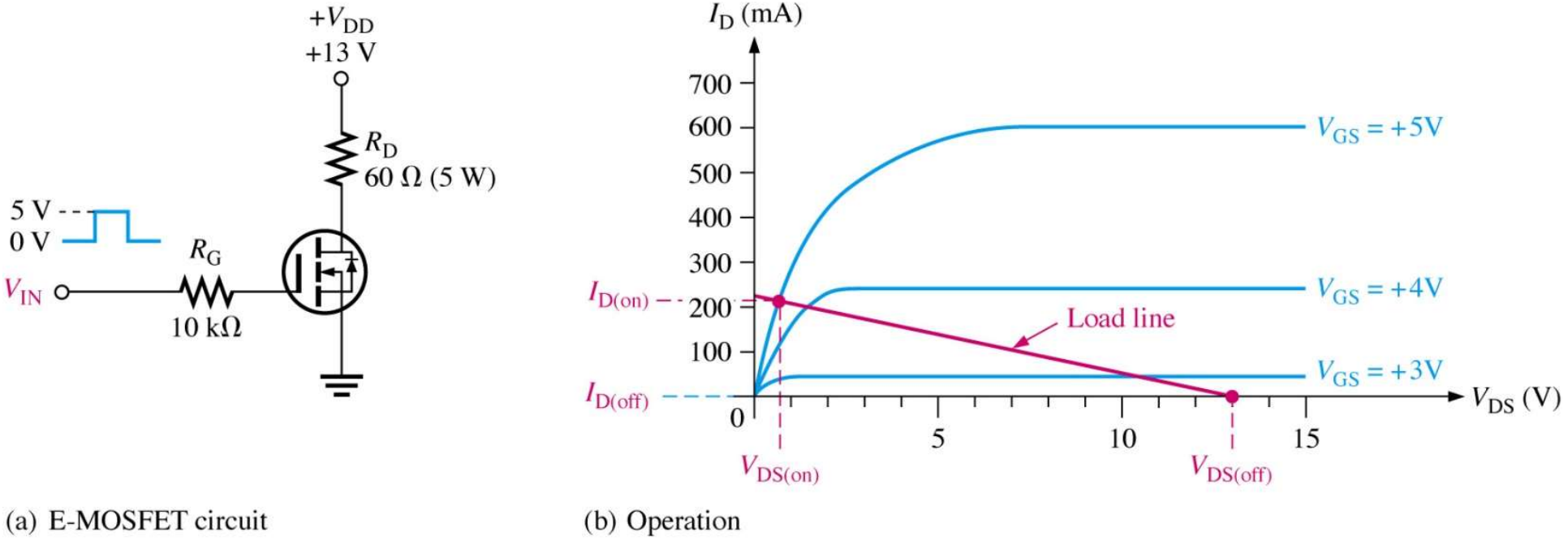


Solid-State Relays (SSR)



The Digital Switch

- **Discrete MOSFET Switches**

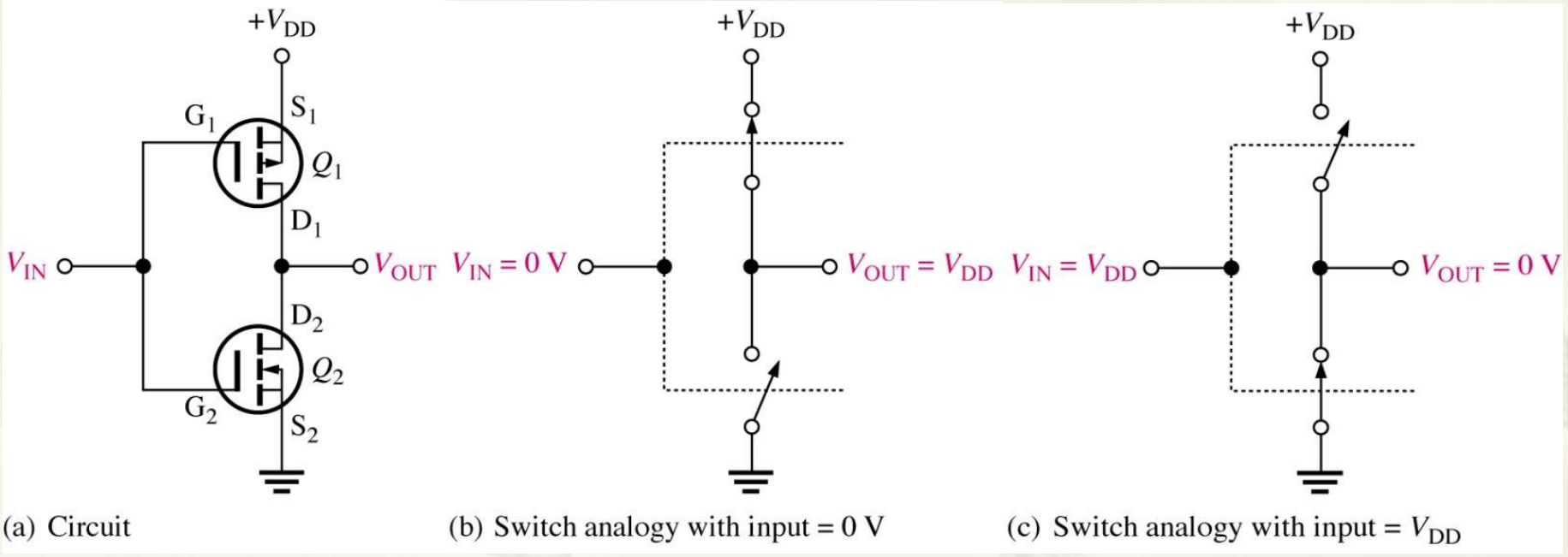


(a) E-MOSFET circuit

(b) Operation

The Digital Switch

- **IC Switching Circuit**
  - CMOS inverter circuit



(a) Circuit (b) Switch analogy with input = 0 V (c) Switch analogy with input = V_{DD}

KEY OBJECTIVES

1. Describe the classifications for field-effect transistors (FETs).
2. Describe the characteristic curve of a JFET and explain common JFET parameters.
3. Explain how to bias a JFET using self-bias and voltage-divider bias.
4. Describe the characteristics of a MOSFET.
5. Explain how to bias a D-MOSFET using zero bias, voltage-divider bias, or current-source bias and an E-MOSFET using voltage-divider bias.
6. Analyze common-source and common-drain FET amplifiers to determine basic operational parameters including gain and input resistance.
7. Discuss two types of FET switching circuits.